Infections & Inflammation

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 26	Infections of the Urinary Tract
Ch 27	Prostatitis, Orchitis & Pain Conditions
Ch 28	Interstitial Cystitis / Bladder Pain Syndrome
Ch 29	Sexually Transmitted Diseases
Ch 30	Cutaneous Diseases of the External Genitalia
Ch 31	Tuberculosis & Parasitic Infections
★	Putting it together: infection decisions

26 Infections of the Urinary Tract

1 · Definitions and classification

- **UTI**: urothelial inflammatory response to bacterial invasion, usually with bacteriuria + pyuria. Urine is **not sterile**; bacteriuria may even protect.
- **Bacteriuria** (colonisation, infection or contamination) and **pyuria** (infection or inflammation: stone, foreign body) are not synonymous with UTI. Bacteriuria without pyuria → colonisation. **Sterile pyuria** → evaluate (TB *→ Ch 31*, stone, urothelial injury, interstitial nephritis).
- **Cystitis**: dysuria, frequency, urgency ± suprapubic pain. Mimics: urethritis, vaginitis, IC/PBS *→ Ch 28*, bladder carcinoma, calculi.
- **Acute pyelonephritis**: chills, fever, flank pain + bacteriuria/pyuria; a clinical diagnosis, not perinephric stranding alone. Flank symptoms may be absent in SCI.
- **Chronic pyelonephritis**: focal coarse cortical scar over a **distorted calyx** (identical to VUR or calyceal-stone scarring). Uniform global thinning suggests obstruction + infection.

	Uncomplicated UTI	Complicated UTI
Host	Healthy, non-pregnant woman	Compromised host and/or virulent or resistant organism
Tract	Structurally and functionally normal	Structural or functional abnormality, catheter
Organism	Eradicated by short oral course	Often MDR → culture mandatory

- **Complicating factors**: anatomical/functional abnormality (BPH, diverticula, ureteric obstruction, NGB, BOO, concentrating defects), **male sex**, pregnancy, elderly, DM, immunosuppression, childhood UTI, recent antibiotics, catheter, instrumentation, hospital-acquired, symptoms >7 days. MCQ Male sex alone makes UTI complicated.
- FDA 2018 complicated UTI: pyuria + pathogen in urine/blood + local or systemic features (fever >38 °C, flank pain, CVA tenderness) with abnormality or catheter.

Term	Definition	Discriminator
Isolated	First, or one remote UTI	—
Unresolved	Therapy failed; same organism	Never cleared
Reinfection	New inoculation (~**70%** of recurrences, often different organism)	Cleared, then new event
Bacterial persistence (relapse)	Same organism from a focus (stone, prostate)	Focus remains; recurs quickly

- **Prophylaxis** prevents reinfection (sterile tract); **suppression** prevents growth from an ineradicable focus.
- **CAUTI**: commonest cause of secondary bacteraemia; ~**80%** of nosocomial UTIs.
- Recurrence risk rises with number of previous UTIs, falls with time since the last. Uncomplicated recurrent UTI does not cause scarring, hypertension or renal failure.

2 · Pathogenesis

Infection = virulence × inoculum versus host defence.

Route	Frequency	Features
Ascending	Commonest	Bowel/skin flora → introitus → urethra → bladder → ureter. Aided by faecal soiling, spermicide, catheters, antiperistaltic factors (endotoxin, pregnancy, obstruction). Reflux not required.
Haematogenous	Uncommon	*S. aureus*, *Candida*; enhanced by obstruction
Lymphatic	Rare	Severe bowel infection, retroperitoneal abscess

Organisms

- ***E. coli***: ~**85%** community, ~50% hospital UTIs; share falls from ~¾ in uncomplicated cystitis to <½ in complicated UTI/urosepsis.
- ***S. saprophyticus***: ~10% of cystitis in young sexually active women; only in uncomplicated cystitis.
- Nosocomial/complicated: *Klebsiella*, *Enterobacter*, *Serratia*, *Pseudomonas*, *E. faecalis*. Culture-negative chronic symptoms: *Ureaplasma*, *M. hominis*.
- UPEC mostly phylogroup B2. **ST131 (O25b:H4)**: MDR clone; ESBL; hallmark fluoroquinolone resistance.
- MCQ *Enterococcus* is a Gram-positive coccus, not an Enterobacteriaceae.
- **Anaerobes**: rare in cystitis, but in most scrotal, prostatic and perinephric abscesses in men and in enteric fistulae. Suspect when microscopy shows organisms but aerobic culture is sterile.
- Urinary TB: haematuria/sterile pyuria; **PCR** has replaced acid-fast staining.

Virulence — fimbriae

- UPEC virulence genes on pathogenicity-associated islands (PAIs): adhesins, toxins (HlyA, Sat), capsule, flagella, iron acquisition. Adherence is the essential first step.

	Type 1 pili (mannose-sensitive)	P pili (mannose-resistant)
Gene / adhesin	*fim*; **FimH** tip adhesin	*pap* operon on a PAI
Receptor	Mannosylated **uroplakins Ia/Ib**	Glycolipid (P blood-group) receptors in kidney
Tropism	Bladder → cystitis	Kidney → **pyelonephritis** (~**90%** of pyelonephritis strains)

- P-pilus frequency: pyelonephritis > cystitis > ASB > faecal isolates. **Phase variation**: switching piliated ↔ non-piliated; type 1 pili promote phagocytosis in suspension, so P pili take over in the kidney.

Intracellular cycle (why cystitis recurs)

> **ALGORITHM**
> FimH binds uroplakin → umbrella-cell invasion (15 min–1 h) → **intracellular bacterial community (IBC)**, biofilm-like (4–16 h) → filamentation, escape, reinvasion (16–24 h) → exfoliation → **quiescent intracellular reservoirs (QIRs)** (from ~2 weeks)

- QIRs are inert and silent; urothelial disruption reactivates them → same-strain recurrence without a urological focus. Biofilm blocks antibiotics and neutrophils.

Host defences and susceptibility

- **Introital receptivity**: greater *E. coli* adherence to vaginal cells in recurrent-cystitis women (genetic; Lewis phenotypes).
- Lactobacilli are the barrier; disturbed by antimicrobials, spermicides, oestrogen loss (pH rises). **Uromodulin (Tamm–Horsfall)**, the most abundant urinary protein, competes with uroplakins for type 1 pili.
- Bladder emptying is the key defence; **obstruction** makes infection life-threatening.
- **VUR**: in children high-grade VUR + UTI → scarring, renal failure; in adults VUR alone harmless without high pressure/stasis + UTI.
- **DM**: more pyelonephritis, abscess, emphysematous infection, fungi, papillary necrosis. **HIV**: UTI ~5×; CD4 <200 → atypical organisms; TMP-SMX Pneumocystis prophylaxis also reduces UTI. **Transplant**: >⅓ get UTI; graft pyelonephritis predicts decline. **SCI**: highest morbidity of any bacteriuric group.

Renal papillary necrosis

- Causes: **DM, obstruction/pyelonephritis, analgesic abuse, sickle cell**; also rejection, cirrhosis, dehydration, renal vein thrombosis, candidiasis. Sloughed papilla → ureteric obstruction; retained papillae calcify → nidus. Treat: antibiotics + early relief of obstruction.

3 · Evaluation and diagnosis

Syndrome	Features
Cystitis	Dysuria, frequency, urgency, suprapubic pain, haematuria; afebrile
Pyelonephritis	Fever, chills, flank pain, CVA tenderness ± vomiting; LUTS may be absent
Fistula	Pneumaturia, faecaluria
Renal/perirenal abscess	Indolent fever, flank mass
Catheter	Usually ASB; bacteraemia can develop fast

	Cystitis	Vaginitis	Urethritis
Onset	Acute	Subacute	Subacute
Frequency/urgency	Prominent	Absent; no haematuria or suprapubic pain	Less marked; no fever
Clues / causes	*E. coli*	Discharge, odour, new partner; HSV, gonorrhoea, *Chlamydia*, *Trichomonas*, yeast, BV	Discharge; initial-stream pyuria in men; *N. gonorrhoeae*, *Chlamydia*, HSV → *Ch 29*

- Dysuria + frequency without discharge → ~**90%** probability of UTI; haematuria + nitrite → ≥90%. Dysuria is the best single symptom.
- **Rule**: painless gross haematuria, or microhaematuria with negative culture or persisting after infection clears → full upper + lower tract evaluation.

Urine collection

- Midstream default (contamination common in women); in-and-out catheter more accurate, halves unnecessary prescribing; suprapubic aspirate most accurate.
- Indwelling catheter: replace and sample the new one. Stoma: catheterise, never the bag.

Urinalysis and culture

- **Leucocyte esterase** = pyuria. **Nitrite**: Enterobacteriaceae; negative with Gram-positives, *Pseudomonas*, *Acinetobacter*, low-nitrate diet, short dwell. High specificity, low sensitivity; best rule-in. Dipsticks better at **ruling out**; negative automated urinalysis excludes UTI.
- Microscopy: pyuria = ≥**5–10 WBC/high-power field (HPF)**; squamous cells = contamination. No pyuria → question diagnosis; culture-positive without pyuria → repeat by catheter.

Threshold	Use
≥10^5 CFU/mL	Kass (originally pyelonephritis); ASB definition
≥**10^2 CFU/mL** of a known pathogen	Symptomatic women with cystitis; catheter specimens

- Low counts in cystitis: frequent voiding washes bacteria out before 10^5.
- "I" susceptibility may still work: urine levels are often several hundred × serum. All common oral agents reach inhibitory urine levels except macrolides.
- **Microbiome** (sequencing, EQUC): "no growth" ≠ sterile, organism ≠ infection; sequencing gives no susceptibilities; decide on symptoms.
- Localisation: **ureteric catheterisation (Stamey)** for one kidney, ectopic ureter or stump. **Stone culture** predicts post-procedure sepsis/SIRS better than bladder urine.

Imaging

Imaging asks only: obstruction/collection to drain, or a focus of persistence? Most UTIs need none.

- **Indications in pyelonephritis**: possible obstruction (stone, stricture, tumour), papillary necrosis risk, prior reconstruction, **no response after 72 h**, DM, polycystic kidneys on dialysis, NGB, unusual organisms (TB, fungi, urea-splitters). Also most men, compromised hosts, rapid recurrence.

Question	Best test
Stone	Non-contrast CT
Abscess, AFBN, drainage planning	Contrast CT
Hydronephrosis, pyonephrosis, PVR	US
Extrarenal extension	MRI
Urethral diverticulum	Pelvic MRI with contrast
VUR; non-refluxing stump	VCUG (febrile UTI in women, childhood VUR, recurrent adult pyelonephritis)
Split function, scarring	DMSA

Correctable causes of bacterial persistence (Box 26.6)
Infection stones; chronic bacterial prostatitis; unilateral infected atrophic kidney; duplication/ectopic ureter; foreign body; urethral diverticulum/periurethral glands; unilateral medullary sponge kidney; non-refluxing infected ureteric stump; infected urachal cyst; infected communicating calyceal cyst; papillary necrosis; perivesical abscess with fistula. **Cure only by removal or drainage.**

4 · Antimicrobial principles

- Cystitis: **urinary** levels matter. Parenchymal infection/bacteraemia: **serum and tissue** levels. Renal insufficiency and obstruction lower urine levels; adjust renally cleared drugs.
- Cystitis goal = symptom relief; NSAIDs alone → longer symptoms, more pyelonephritis.
- **Collateral damage** (resistance, MDR colonisation, *C. difficile*, candidiasis), mainly broad cephalosporins and quinolones.
- P1 Antimicrobial resistance; ESBL producing enterobacterial UTI — *intrinsic / chromosomal / plasmid resistance; ESBL, AmpC, KPC and their treatment options*
- Plasmid resistance is transferred in **bowel** flora; one drug selects multi-resistance, so post-antibiotic UTIs are often MDR. Prior quinolone use is the strongest determinant of quinolone resistance.
- **Stewardship**: narrow spectrum, shortest course, culture before treating recurrent/complicated UTI, local antibiograms.

Agents

Class	Action	Resistance
β-lactams	Cell wall	β-lactamase, altered PBP, porins
Fosfomycin	Cell wall (different step) → little cross-resistance	Transporter loss
Vancomycin	Cell wall	Altered peptidoglycan
Aminoglycosides	Ribosome, bactericidal	Modifying enzymes
Quinolones	DNA gyrase	Gyrase mutation, efflux, porins
TMP-SMX	Folate antagonism	Enterococci scavenge folate (fail even if "sensitive")
Nitrofurantoin	Multiple enzymes	Very low

- **Anti-pseudomonal**: antipseudomonal penicillins, **ceftazidime** (not ceftriaxone), aztreonam, aminoglycosides, fluoroquinolones. **Enterococcal**: aminopenicillins, piperacillin, nitrofurantoin, fosfomycin; **never cephalosporins**. Aztreonam, pivmecillinam: no Gram-positive cover; vancomycin: no Gram-negative. Ertapenem misses *Pseudomonas*, *Enterococcus*; carbapenems miss *E. faecium*, MRSA.

Agent	Key points
Nitrofurantoin	Urine-only → useless for pyelonephritis, prostate, bacteraemia. Flora-sparing. Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis. Beers 2015: avoid if CrCl < **30 mL/min** (older 50–60); no long-term suppression in elderly.
TMP-SMX	Good tissue levels; misses *Enterococcus*, *Pseudomonas*. Commonest antibiotic cause of SJS/TEN; potentiates warfarin.
Fosfomycin	3 g single dose; active against ESBL and VRE → oral MDR cystitis option.
Fluoroquinolones	Broad incl. *Pseudomonas*. Adjust at CrCl <30. FDA warnings: tendon rupture (age >60, steroids, renal failure), neuropathy, dysglycaemia, psychiatric, aortic dissection; **2016: reserve for no alternative in uncomplicated UTI**. Avoid in children and pregnancy. Separate antacids/iron ≥2 h; theophylline, warfarin.
Cephalosporins	Poor on enterococci; 3rd generation most reliable β-lactam for Gram-negatives. Cystitis only if first-line agents unusable.

Agent	Key points
Carbapenems	Stable to ESBL and AmpC; penicillin cross-sensitivity <1%.
Aminoglycosides	Once daily 5–7 mg/kg (concentration-dependent killing, post-antibiotic effect, less nephrotoxicity); nomogram level 6–14 h post-dose sets interval 24/36/48 h. Monitor oto/nephrotoxicity.
Aztreonam	Aminoglycoside-like spectrum without nephrotoxicity; penicillin allergy, renal impairment.
Pivmecillinam	~2% *E. coli* resistance; Nordic first-line; not in USA.

KEY POINT

Narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis; broad tissue-penetrating agents (quinolones, cephalosporins, carbapenems) are kept for invasive infection. Resistance is made in the bowel, so breadth, duration and repetition matter more than MIC.

C. difficile infection (CDI)

- Culprits: cephalosporins, clindamycin, carbapenems, quinolones, TMP-SMX. Diagnosis: ≥3 unformed stools/24 h + toxigenic *C. difficile*/toxin.

Episode (IDSA/SHEA 2021)	Treatment
Initial	**Fidaxomicin** 200 mg twice daily × 10 d; alternative oral vancomycin 125 mg four times daily × 10 d; metronidazole only if non-severe and others unavailable
First recurrence	Fidaxomicin or tapered vancomycin ± **bezlotoxumab** (anti-toxin B)
≥2 recurrences	As above, or **faecal microbiota transplantation**
Fulminant	Oral/rectal vancomycin 500 mg four times daily + IV metronidazole

5 · Asymptomatic bacteriuria

Group (IDSA)	Specimen	Threshold
Women	**Two** consecutive voided, same strain	≥10^5 CFU/mL
Men	One voided	≥10^5 CFU/mL
Either sex	One catheterised	≥**10^2** CFU/mL

- Prevalence: premenopausal women 1–5%, long-term care 25–50%, long-term catheter **100%**.
- Untreated ASB causes no hypertension, renal failure, cancer or recurrent UTI. Treating it → more symptomatic UTI, MDR, adverse effects; ASB strains may protect (bacterial interference).

Do NOT screen or treat (IDSA 2019)	Screen and treat
Premenopausal non-pregnant women; DM; elderly (community/institution); SCI; indwelling catheters; ASB with pyuria; children; renal transplant >1 month; other solid-organ transplant; **non-urological surgery (incl. arthroplasty, cardiac)**	**Pregnancy**; **urological procedures with anticipated mucosal bleeding** (TUR, ureteroscopy, PCNL). No consensus: neutropenia.

KEY POINT
ASB is treated only where bacteria are about to reach blood or tissue: pregnancy (hydroureter → pyelonephritis) and mucosa-breaching urological surgery. Elsewhere the resident strain is harmless or protective.

6 · Cystitis

Acute uncomplicated cystitis in women

- Risk factors by defence defeated: **washout** (obstruction, NGB, dehydration); **flora/binding** (intercourse, spermicide, oestrogen loss, antibiotics); **ascent** (catheter, incontinence, residual).
- Diagnosis: acute dysuria + change from baseline voiding. Fever/flank pain or no oral route → treat as pyelonephritis.

Tier (IDSA/ESCMID)	Regimen	Caveat
First-line	**Nitrofurantoin** 100 mg twice daily × **5 d**	Not if pyelonephritis suspected
First-line	**TMP-SMX** 160/800 mg twice daily × **3 d**	Avoid if local resistance > **20%** or used in past 3 months
First-line	**Fosfomycin** 3 g single dose	Slightly lower efficacy
First-line	**Pivmecillinam** 400 mg twice daily × 3–7 d	Not in North America
Second-line	Fluoroquinolone × 3 d	Ecology and toxicity; last resort (FDA 2016)
Second-line	β-lactam (amoxicillin–clavulanate, cefpodoxime, cefadroxil) × 3–7 d	Lower efficacy; never amoxicillin alone

3 days beats single dose and equals 7 days (7 if complicating factors). No post-treatment culture if asymptomatic.

Cystitis in men

- Complicated; about half have an abnormality (incomplete emptying) → evaluate; consider prostatitis → *Ch 27*.
- Afebrile men: **7 days = 14 days** (Drekonja). EAU: TMP-SMX 160/800 twice daily × 7 d; quinolone per local susceptibility.

Complicated UTI

Category (CUA)	Examples
Anatomical	Cystocele, fistula, diverticulum
Iatrogenic	Nosocomial, surgery, catheter
Obstruction	BOO, PUJ obstruction, ureteric stricture
Voiding dysfunction	Neurological disease, VUR, incontinence, high PVR
Other	Pregnancy, stones, DM, immunosuppression

Always culture; correct the abnormality; **10–14 days** culture-directed; IV → oral once afebrile; re-culture if no response.

Emphysematous cystitis

- Bladder-wall gas; elderly diabetic woman. *E. coli* ~60%, *Klebsiella*. Abdominal pain (~80%), gross haematuria; fever only 30–50%. Plain film: curvilinear lucency; **CT confirms**, excludes fistula.
- ~**90%** medical: broad Gram-negative cover, catheter drainage, glucose control; cystectomy/debridement rarely.

Unresolved bacteriuria

Cause (descending frequency)	On-treatment culture	Action
1. Resistant from outset (recent antibiotics)	Resistant	Change drug; no further work-up
2. Emergent resistance (underdosing)	Same species, now resistant	
3. Unsuspected second pathogen	New species present from start	
4. Rapid reinfection, new resistant species	New species → suspect enterovesical fistula	
5–7. Azotaemia; papillary necrosis; giant staghorn	Same species, still **sensitive**	Renal function + imaging
8. Non-compliance, Munchausen	—	—

7 · Recurrent UTI

- Definition: **2 in 6 months or ≥3 in 12 months**, culture-documented; ~¼ of young women recur within 6 months.
- Reinfection = different organism or same organism >2 weeks after treatment (commoner; faecal flora or QIRs). In men, reinfection also suggests an anatomical abnormality.
- Risk factors: intercourse, new partner, spermicide, first-degree female relative, premenopausal UTI, menopause, recent antibiotics; incontinence, cystocele, raised PVR.
- Evaluation: history (pneumaturia, faecaluria, diverticulitis, radiotherapy), pelvic exam, **culture each episode**, PVR ± flow. **Imaging and cystoscopy not routine** (AUA, EAU, CUA).

Indications for imaging ± cystoscopy in recurrent UTI (CUA)
Prior urinary surgery/trauma; stones; gross haematuria or persisting microhaematuria; obstructive symptoms, low flow, high PVR; urea-splitters; prior abdominopelvic malignancy; persistence despite sensitivity-based therapy; DM/immunocompromise; pneumaturia, faecaluria, anaerobes, diverticulitis; repeated pyelonephritis. MCQ Recurrence frequency alone is **not** an indication.

- **Struvite**: urease (*Proteus mirabilis*) → magnesium ammonium phosphate; bacteria inside the stone. AUA: remove infection stones completely in fit patients; unfit → acetohydroxamic acid + suppression.

Prevention (women, AUA 2019)

> **ALGORITHM**
> Confirm recurrent UTI (symptoms + culture) → same organism at short intervals? → evaluate for persistence (upper tract imaging + cystoscopy), remove nidus → reinfection: behavioural (≥1.5 L water, frequent and postcoital voiding, stop spermicide) → cranberry ≥36 mg PAC → coital: **postcoital antibiotic**; postmenopausal: **vaginal oestrogen**; neither: **methenamine hippurate** (normal renal function) → continuous prophylaxis last

Measure	Mechanism	Evidence / use
Water ≥1.5 L/day	Washout	RCT (premenopausal): 1.7 vs 3.2 episodes/yr
Vaginal oestrogen	Restores lactobacilli, lowers pH	Strongest non-antibiotic evidence; ~90% reduction (Raz–Stamm); benefit after ≥12 weeks. **Oral oestrogen ineffective.** No excess breast cancer/cardiovascular risk; breast cancer survivors with oncologist agreement.
Cranberry ≥36 mg PAC (twice daily)	Type A PAC blocks **P-fimbrial** adhesion	Cochrane 2023 favourable; safe with warfarin
Methenamine hippurate 1 g twice daily	**Formaldehyde in acid urine (pH <6)**	Non-inferior to daily prophylaxis. Fails with urease producers, NGB/anatomical abnormality. Contraindicated CrCl <50, liver disease; antacids, carbonic anhydrase inhibitors, citrus reduce effect.
D-mannose	Inhibits **FimH**	Cochrane 2022 inconclusive
Immunostimulants (OM-89 oral, Urovac vaginal, Uromune MV140 sublingual)	Bacterial lysates	Some benefit; small biased trials
Probiotics, oral oestrogen, vitamin C	—	Not effective

Antibiotic strategy	Evidence	Position
Continuous low dose 6–12 months (nitrofurantoin, TMP, TMP-SMX, cephalexin)	Works only while taken; MDR	AUA "may prescribe"; EAU: last resort
Postcoital single dose	Effective; preferred for coital UTI	Acceptable
Self-start	Old data; overuse	Prefer analgesic + sample + observe, then culture-directed

KEY POINT

Recurrent UTI in women is reinfection in a receptive host, not a hidden focus. Investigate only for features of persistence or complication; prevention targets flora, pH, adhesion and washout rather than repeatedly sterilising urine.

8 · Kidney infections

Acute pyelonephritis

- Abrupt chills, fever ≥38 °C, flank/CVA pain; LUTS often absent; may mimic GI disease. Labs: leucocytosis, CRP; **WBC casts** = renal origin. Blood cultures if toxic, admitted or pregnant.
- Usually P-fimbriated *E. coli*; suspect resistance after recurrent UTI, hospitalisation, catheter, instrumentation.

Setting (IDSA)	Regimen	Days
Outpatient, no vomiting	Ciprofloxacin 500 mg twice daily if community quinolone resistance ≤ **10%**; levofloxacin 750 mg daily; TMP-SMX DS twice daily only if known sensitive. ± one IV dose ceftriaxone 1 g or aminoglycoside.	Cipro 7; levo 5; TMP-SMX 14
Inpatient, severe/possible sepsis	Ampicillin + gentamicin; levofloxacin; ceftriaxone 1 g daily; carbapenem (ESBL risk). Gram-positive cocci → ampicillin–sulbactam ± aminoglycoside.	10–14 total

Setting (IDSA)	Regimen	Days
Pregnancy	Ampicillin + gentamicin, aztreonam, ceftriaxone → oral cephalexin. **No quinolone.**	10–14

- IV if high fever, leucocytosis, vomiting, dehydration, sepsis, complicated, outpatient failure. Positive blood cultures → IV until stable; negative → 2–3 days IV then oral to 10–14 days.
- **Obstruction** → urgent decompression: **stent or percutaneous nephrostomy**. Solitary obstructed kidney → ARF; dose by GFR.

ALGORITHM

Pyelonephritis → no sepsis/vomiting: culture ± US → oral outpatient 7–10 d | sepsis: blood + urine cultures, imaging → IV inpatient → improved by **72 h**: oral step-down | not improved: repeat cultures + **CT** → drain obstruction or abscess

KEY POINT

Urine sterilises within hours but fever may lag for days. Fever beyond 72 h means obstruction or an undrained collection until CT proves otherwise. Antibiotics treat the infection; drainage treats the complication.

Acute focal bacterial nephritis (AFBN)

- Lobar leucocyte infiltrate between pyelonephritis and abscess; half diabetic, half bacteraemic. CT: wedge-shaped, poorly marginated, no wall or liquefaction, seen only after contrast (abscess: round, liquid centre, **ring sign**).
- IV antibiotics ≥7 days then 7 days oral. Non-response → exclude abscess, obstruction, RCC, renal vein thrombosis.

Emphysematous pyelonephritis (EPN)

- P1 P3 EPN. Prognostic factors & conserv. mgt.; Prognostic indicators in emphysematous pyelonephritis *— definition, CO_2 mechanism, CT choice, Wan patterns, Huang–Tseng risk factors, drainage-first management*
- Mostly **diabetic women**; triad fever, vomiting, flank pain; urine culture almost always positive. Non-diabetics usually obstructed, less extensive.

CT pattern	Prognosis
Streaky/mottled gas, **no fluid**	50–60% mortality
Fluid with bubbly/loculated gas, or gas in collecting system	<20% mortality

Huang-Tseng class †	Extent of gas	Management †
1	Collecting system only (emphysematous pyelitis)	Antibiotics + PCD/stent
2	Parenchyma, no extrarenal extension	Antibiotics + PCD
3A	Perinephric space	PCD if <2 risk factors; nephrectomy if ≥2 or PCD fails
3B	Pararenal space	
4	Bilateral or solitary kidney	PCD (nephron-sparing)

† Risk factors: thrombocytopenia, AKI, altered consciousness, shock. Assess contralateral function with a nuclear scan. Emphysematous pyelitis is often non-diabetic and milder.

KEY POINT
Fluid is a good sign and streaky gas a bad one: a contained collection can be drained; diffuse gas without fluid means parenchyma is necrosing faster than the host can wall it off.

Renal and perinephric abscess

- P1 Renal and perinephric abscess — *haematogenous vs ascending, perinephric causes, insidious course, CT, empirical antibiotics, size-based drainage (≤3 / 3–5 / >5 cm), nephrectomy*
- Renal abscess: ≤**3 cm** antibiotics; >**5 cm** percutaneous drainage; any size if immunocompromised/non-responding. Look for skin source 1–8 weeks earlier; aspirate if abscess vs cystic tumour uncertain.
- Perinephric: lower drainage threshold (>3 cm). Rupture through Gerota → psoas abscess (flexed, externally rotated hip). Only ⅓ diagnosed at presentation; delay causes nearly all deaths.

Infected hydronephrosis and pyonephrosis

- **Pyonephrosis**: infected hydronephrosis with suppurative destruction and near-total loss of function; very ill; **bacteriuria may be absent** if ureter completely obstructed. US: fluid–debris level.
- Antibiotics + **immediate drainage** (stent or nephrostomy) → treat cause once stable; nephrectomy if non-functioning.

XGP and malakoplakia

- P1 P2 Xanthogranulomatous pyelonephritis; Malakoplakia — *stone–obstruction–Proteus triad, bear-paw CT, RCC mimic, complete nephrectomy; Michaelis–Gutmann bodies, intracellular antibiotics + bethanechol/vitamin C*
- XGP extras: hepatic dysfunction (up to half); renal failure uncommon (unilateral). Differential: lymphoma (bilateral, no stones).

	XGP	Malakoplakia
Cell	Foamy lipid-laden macrophages	Von Hansemann cells, Michaelis–Gutmann bodies
Stones/obstruction	Usually	Absent
Organism	*Proteus, E. coli*	*E. coli*
Laterality	Unilateral	Uni-/multifocal, may be bilateral
Treatment	Nephrectomy	Intracellular antibiotics ± nephrectomy

KEY POINT
Across renal infections the axis is diffuse inflammation versus contained collection. Pyelonephritis and AFBN need antibiotics; pyonephrosis and large or obstructed abscesses need drainage; when the kidney itself is the infected tissue (XGP, diffuse EPN), the answer is nephrectomy.

9 · Fournier gangrene and periurethral abscess

- **Fournier gangrene**: necrotising fasciitis of genitalia, perineum and/or perianal region. Source found in ~**95%**; age 50–79; male:female ~10:1.
- Sources: **colorectal** (perianal abscess), **urogenital** (stricture with extravasation, instrumentation, catheter), **dermal** (hidradenitis, balanitis). Risk: **DM**, obesity, alcoholism, immunosuppression, trauma, paraphimosis, surgery.

- Polymicrobial **aerobic–anaerobic synergy** → obliterative endarteritis, thrombosis → necrosis. Spread: Buck's → dartos → Colles' → Scarpa's.
- Pain and oedema out of proportion → crepitus → dusky skin → gangrene. **Skin underestimates fascial extent**; systemic toxicity out of proportion is the warning.
- Labs: anaemia, raised creatinine, hyponatraemia, **hypocalcaemia** (lipase-released fatty acids chelate calcium). CT best for fascial gas, but must not delay surgery.

Score	Components	Use
LRINEC	CRP, WBC, haemoglobin, sodium, creatinine, glucose	≥ **6** suspicious; ≥ **8** strongly predictive
FGSI †	Temperature, HR, RR, Na, K, creatinine, haematocrit, WBC, bicarbonate (each 0–4)	>9 predicts mortality

ALGORITHM
Suspicion → resuscitation + broad IV antibiotics (piperacillin–tazobactam; or carbapenem/vancomycin + clindamycin or metronidazole) → **immediate radical debridement** to bleeding fascia, leave open → **re-look at 24–48 h** → suprapubic diversion (urethral source) / colostomy (rectal source) → hyperbaric oxygen (adjunct) → reconstruction

- **Orchidectomy almost never needed**: testicular artery independent of scrotal skin supply.
- Mortality ~20–40%; worse with liver failure, alcoholism, DM, age, cardiac/renal failure, coagulopathy.
- **Periurethral abscess** (stricture, catheter): debridement/drainage + suprapubic diversion + antibiotics.

10 · Urosepsis †

Term (Sepsis-3)	Definition
Sepsis	Organ dysfunction from dysregulated response to infection: SOFA rise ≥ **2**
qSOFA	≥2 of: RR ≥22/min, altered mentation, systolic BP ≤100 mmHg
Septic shock	Vasopressors for MAP ≥65 mmHg **and** lactate >2 mmol/L despite fluids

ALGORITHM
Hour-1 bundle: lactate → blood + urine cultures → broad IV antibiotic within 1 h → 30 mL/kg crystalloid if hypotension or lactate ≥4 → noradrenaline for MAP ≥65 → **source control**: decompress (stent or nephrostomy; no definitive stone surgery) → de-escalate by culture

† Not in the source notes. Manipulating infected stones releases endotoxin: drain first, operate later.

11 · Special populations

Pregnancy

- P1 P3 Bacteriuria in pregnancy; Baceriuria and use of antibiotic policy in pregnancy — *prevalence unchanged but ascends, culture screen at 12–16 weeks, 4–7 day course, safe/avoid drug table, pyelonephritis and suppression*
- GFR +30–50% → creatinine 0.7 → **0.5 mg/dL**, so 0.8–0.9 may mean impairment; proteinuria abnormal >300 mg/24 h. Kidney +1 cm.

- Also safe: aztreonam. Avoid chloramphenicol ("grey baby"). Severe/MDR pyelonephritis → carbapenem.

Elderly

- **Do not screen or treat ASB**. Bacteriuria + confusion ≠ UTI: without urinary symptoms hydrate and reassess. Do not culture for non-specific symptoms or to document clearance.
- Pyuria is common, but its **absence excludes** bacteriuria.
- **Revised McGeer** (surveillance): non-catheterised need **localising** signs + ≥10^5 CFU/mL of ≤2 species; catheterised may qualify on **systemic** signs + ≥10^5.
- Treat 3–6 days uncomplicated, 10–14 if febrile; watch renal dosing, quinolone toxicity.

Catheter-associated UTI

- P1 Catheter Associated Urinary Tract Infections (CAUTI) — *definition, 5–10% per catheter-day, extra-/intraluminal routes, biofilm, prevention bundle, change catheter, treat only symptomatic*
- Catheters >30 days: 4–5 species. "6-C" bundle: Consider, Connect, Clean, Closed, Call, Culture only if indicated. Silicone = latex. Stop antibiotics within 48 h of resolution.

Spinal cord injury

- All UTIs complicated; commonest cause of fever. Symptoms atypical: fever, spasticity, leakage, autonomic dysreflexia †, cloudy urine.
- **CIC (Lapides 1972)** has the fewest long-term complications; indwelling urethral catheter the most. Suprapubic catheter delays but does not reduce bacteriuria.

Drainage (NIDRR)	Significant bacteriuria
Indwelling urethral or suprapubic catheter	Any detectable count
Intermittent catheterisation	≥10^2 CFU/mL
Catheter-free men (clean voided)	≥10^4 CFU/mL

- No ASB treatment. Afebrile: 5–7 days (recurrent 7–10); febrile: admit, parenteral, **14 days**. Change the catheter. No response in 24–48 h → image. Post-treatment culture only for urea-splitters.
- Prophylaxis and methenamine ineffective in NGB (prophylaxis doubles resistance). Recurrent UTI → imaging, UDS, lower storage pressure, review CIC.
- Catheter > **10 years** → bladder cancer (often squamous †) → surveillance.

Diabetes, renal transplant

- DM: no ASB treatment; low threshold to image pyelonephritis (EPN, papillary necrosis, abscess).
- Transplant: no ASB screening >1–2 months. Uncomplicated UTI 5–10 days (not 3); pyelonephritis 14–21 days. Stents raise UTI by ~half → remove within **4 weeks**. Recurrent UTI → PVR, imaging, VCUG, UDS.

12 · Fungal UTI

- P1 Fungal UTI — *Candida risk factors, asymptomatic candiduria (remove catheter; treat only neutropenia, VLBW infants, pre-procedure), fluconazole 2 weeks, amphotericin for resistant species, fungus ball drainage, IDSA vs AUA pre-procedure*
- AUA BPS 2019: antifungal prophylaxis for high-pressure endoscopy (TURBT, TURP, therapeutic ureteroscopy) and open tract entry; preoperative fungal culture + 5–7 days

periprocedural antifungal for fungal balls; longer course with neutropenia + obstruction.

13 · Perioperative antimicrobial prophylaxis

Principle (AUA BPS 2019)	Rule
Indication	Any procedure breaching a tissue barrier, weighing host defence and consequences
Timing	Within **60 min** of incision; vancomycin and quinolones within 120 min
Duration	**Single dose**; ≤ **24 h** only for PCNL, prostheses, special cases
Agent	Local flora per antibiogram
Known infection	Sterilise urine before elective surgery (treatment); infected urine → bacteraemia in ~half

- Host risk factors: age, anomaly, malnutrition, smoking, steroids, immunodeficiency, hardware, infected stones, distant infection, prolonged stay.

Wound class	Examples	AP
I clean	Scrotal/inguinal, RPLND, nephrectomy without entering tract	Not needed (except groin/perineal incision, prosthesis)
II clean-contaminated	Any endoscopy; opening the urinary tract	Single dose
III contaminated	Infected stone PCNL, **transrectal biopsy**, bowel segments	Strongly recommended; anaerobic cover
IV dirty	Abscess, open trauma	Treatment, tailor to cultures

Procedure	AP?	Agent	Duration
Diagnostic cystoscopy / UDS (no risk factors)	No (EAU strongly against)	—	—
Cystoscopy + biopsy/fulguration	Host factors	TMP-SMX, amoxicillin–clavulanate, cephalosporin	Single
TURP / TURBT, laser prostate	All (TURBT: EAU only if high risk)	Cefazolin, TMP-SMX	Single (TURP short course with catheter acceptable)
Transrectal prostate biopsy	All	Fluoroquinolone or cephalosporin ± aminoglycoside; targeted by rectal swab	Single (≥1 day)
Ureteroscopy	All (diagnostic: uncertain)	TMP-SMX, 1st/2nd-gen cephalosporin	Single
PCNL	All	1st/2nd-gen cephalosporin; aminoglycoside + metronidazole	≤24 h
SWL	Only with risk factors (EAU: no)	TMP-SMX, cephalosporin	Single
Open/robotic entering tract	All	Cefazolin	Single

Procedure	AP?	Agent	Duration
Colon segment	All	Cefazolin + metronidazole / cefoxitin / ertapenem + oral neomycin with erythromycin or metronidazole	Single
AUS, penile prosthesis, sacral neuromodulation	All	Aminoglycoside + cephalosporin or vancomycin	≤24 h
Vaginal surgery incl. slings	All	Cefazolin/cefoxitin	Single
Penile (circumcision), clean scrotal	Likely none (scrotal: high-risk only)	Cefazolin	Single

- Aztreonam replaces aminoglycosides in renal impairment. No routine vancomycin.
- **TRUS biopsy**: sepsis 0.3–3.1%, mainly quinolone-resistant *E. coli*; strongest risk = antibiotics within 6 months. Options: augmented, **targeted** (rectal swab), fosfomycin. AUA: alternatives if local resistance >20%. EAU: povidone-iodine rectal preparation.
- **UDS (SUFU 2017)**: AP if neurogenic LUT dysfunction, raised PVR, ASB, immunosuppression, age >70, catheter, recent joint implant; not for DM, prior UTI, valve disease or obesity alone.
- **Catheter/stent removal**: single dose with risk factors; none for chronic catheter changes or long-term CIC. Culture the stent in high-risk patients.
- **Endocarditis** (AHA): no AP solely for prevention; high-risk valve + colonised urine → sterilise urine (amoxicillin for enterococcus).
- **Orthopaedic hardware**: pins, plates, screws never need AP. Joint <2 years or high-risk host + high-bacteraemia procedure (stone manipulation, upper tract endoscopy, bowel, transrectal biopsy) → AP (quinolone, or ampicillin + gentamicin).

KEY POINT

Prophylaxis is justified by wound contamination × host vulnerability. Where neither is raised (diagnostic cystoscopy, sterile-urine SWL, index UDS), ecological cost outweighs benefit; otherwise give one well-timed, flora-matched dose. Anything longer is treatment.

14 · Landmark trials

Trial	Design	Result	Take-home
Raz & Stamm 1993	RCT, intravaginal oestriol vs placebo, postmenopausal recurrent UTI	0.5 vs 5.9 UTIs/patient-year; lactobacilli returned, pH fell	Vaginal oestrogen first-line after menopause
Hooton 2018	RCT, +1.5 L water/day vs none, premenopausal	Cystitis 1.7 vs 3.2/year	Hydration halves recurrences
Cai (ASB in recurrent UTI)	RCT, treat vs observe ASB in young women	Treatment ↑ recurrence and resistant *E. coli*	Do not treat ASB; *E. faecalis* may protect
ALTAR (methenamine)	Methenamine vs daily antibiotic prophylaxis, 12 months	Non-inferior	Non-antibiotic alternative
Drekonja 2021	RCT, afebrile male UTI, 7 vs 14 days ciprofloxacin/TMP-SMX	7 days non-inferior	7 days for afebrile men

27 Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

1 · Prostatitis: definitions and classification

- **Drach (1978)**, built on **Meares–Stamey (1968)** segmented cultures: acute bacterial, chronic bacterial, nonbacterial, prostatodynia. MCQ Replaced by **NIH (1999)**.

NIH category	Entity	Defining feature
I	Acute bacterial prostatitis (ABP)	Acute prostatic UTI, often febrile
II	Chronic bacterial prostatitis (CBP)	Recurrent UTI, **same organism**; well between episodes
IIIA	CP/CPPS, inflammatory	Leucocytes in EPS, VB3 or semen
IIIB	CP/CPPS, non-inflammatory	No leucocytes
IV	Asymptomatic inflammatory	Incidental on biopsy or semen analysis

- I + II = **5–10%**; III = **90–95%**. **IIIA vs IIIB has no clinical significance**; VB3/EPS leucocyte microscopy no longer recommended. The prostate may not be the pain source, hence "CPPS".
- **Category III (NIH CPCRN)**: pelvic pain **≥3 of the previous 6 months**. Exclusions: uropathogens, urogenital cancer, prior radiation/chemotherapy, stricture, neurological bladder disease. MAPP term: urologic chronic pelvic pain syndrome.

Histopathology

- Histological inflammation near-universal (autopsy **>70%**), so it cannot define disease alone.
- **Nickel grading**: location (glandular; periglandular ≤50 µm; stromal >50 µm); extent (focal <10%, multifocal 10–50%, diffuse >50%); grade (1 separate cells; 2 confluent, no destruction; 3 confluent with destruction).
- **Prostatic calculi**: bacterial nidus. **Granulomatous prostatitis** (Epstein–Hutchins: specific, non-specific, post-TURP, allergic): classically after **intravesical BCG → transient benign PSA rise**; also TB.

2 · Category I: acute bacterial prostatitis

- Routes: **ascending urethral** (main); intraprostatic reflux; biopsy; lymphatic, haematogenous (rare). Risks: insertive anal intercourse, phimosis, catheters (CIC up to ⅓), instrumentation/biopsy, obstruction, preceding cystitis/epididymo-orchitis.
- ***E. coli* 65–80%**; *Pseudomonas*, *Proteus*, *Klebsiella*, *Serratia*; *Enterococcus* ≤10%; gonococcus in young men. Post-instrumentation: *Pseudomonas* ~20%, more abscess. Post-biopsy: FQ-resistant, ESBL, bacteraemic. HIV: *S. aureus*, *Salmonella*, mycobacteria.

> **KEY POINT**
> **Why young healthy men get ABP:** the organism, not the host — virulent UPEC that forms **biofilm** more often than cystitis/pyelonephritis strains; biofilm blocks antibiotics, so prostatic infection needs longer, well-penetrating therapy.

- Frequency, dysuria, perineal pain, fever, rigors; retention; sepsis. Post-instrumentation cases are older with more septicaemia.

- Febrile male UTI: **>90%** show PSA and/or prostate volume rise — prostatic involvement is the rule.

Evaluation

- First question: **spontaneous or after manipulation/biopsy?** (decides empirical therapy).
- DRE: tender, boggy prostate; **gentle; massage contraindicated**.
- FBC, MSU culture (positive 60–85%), **blood cultures** if febrile, renal function. NAAT only if discharge. PVR by **ultrasound**, not catheter.
- **No PSA** acutely; if raised, follow to normal (up to **6 months**). Image (TRUS/CT) only for suspected abscess; **no biopsy**.

Treatment

- ~10% progress to II, ~10% to III. **Outpatient only if** no systemic illness, tolerating orals and no retention.
- **EAU**, systemic illness: amoxicillin + aminoglycoside; 2nd-gen cephalosporin + aminoglycoside; or 3rd-gen cephalosporin. Ciprofloxacin only if local resistance **<10%**, fully oral, not hospitalised, or β-lactam anaphylaxis.
- **Post-biopsy febrile prostatitis** (AUA White Paper): **no FQ or TMP-SMX**; **carbapenem**, amikacin or 2nd/3rd-gen cephalosporin.
- Oral step-down: ciprofloxacin 500 mg bd or levofloxacin 500 mg od if susceptible; reculture at 1 week.
- **Duration**: traditionally 4 weeks; **2 weeks** ciprofloxacin equivalent. TB prostatitis ≥ **6 months**. Adjuncts: NSAIDs, α-blockers.
- **Retention**: suprapubic catheter traditional; brief urethral/intermittent acceptable short-term; **suprapubic for long-term**.

Prostatic abscess

- Suspect if high fever, immunosuppression, **no response after 48 h**. Risks: catheters, biopsy, obstruction, CBP, DM, cirrhosis, HIV, dialysis. TRUS hypoechoic; **CT** for spread and planning.

> **ALGORITHM**
> Abscess <1–2 cm → antibiotics → progression → peripheral: US-guided percutaneous drainage (recurrence 15–33%) → large or failed: **transurethral unroofing** (intra-op US to locate) → extraprostatic extension: open drainage (rare)

3 · Category II: chronic bacterial prostatitis

- Same organism harboured in prostate; **asymptomatic between episodes** (CP/CPPS is continuous).
- Organisms: Gram-negatives; *E. faecalis* (dominant Gram-positive). Special culture: TB, *Candida*; *Trichomonas* possible. Biofilm explains poor response.
- **Chlamydia**: debated (urethral contamination); VB3+/VB1− men improved on levofloxacin + azithromycin 4 weeks → some causal evidence.

Localisation test	Specimens	Interpretation
Meares–Stamey 4-glass (previous)	VB1 urethra; VB2 bladder; EPS; VB3 post-massage	Prostatic if EPS/VB3 exceed VB1/VB2
2-glass pre/post-massage (current)	Midstream pre-massage + first 10 mL post-massage	Predicts 4-glass in **>95%**

- No agreed cut-off (some use **10-fold rise**). **No semen culture**. HIV: also anaerobes, fungi, TB.
- DRE nodule → cancer. Recheck haematuria 4–6 weeks after clearance. PVR **>180 mL** raises infection risk.
- Men **<45** with first UTI: imaging normal, but check for **stricture** (up to 41%). Image if DM, stones, voiding/neurological problems, poor response, haematuria.

> KEY POINT
> **Why fluoroquinolones?** Prostatic entry is passive (lipid solubility, ionisation, protein binding, size). FQs are lipid-soluble **zwitterions** concentrating in alkaline CBP fluid and cover Gram-negatives incl. *Pseudomonas*, Gram-positives, atypicals. Tetracyclines, macrolides, trimethoprim also penetrate.

Situation (EAU)	Regimen
First line	FQ (ciprofloxacin = levofloxacin) **4–6 weeks**
FQ-resistant, TMP-SMX-sensitive	TMP-SMX **3 months** (cure 30–50% vs 60–80% FQ)
C. trachomatis	Azithromycin 500 mg daily 3 weeks, or doxycycline 100 mg bd 10 days; azithromycin > ciprofloxacin
T. vaginalis	Metronidazole 500 mg tds 14 days

- Add a macrolide (**penetrates biofilm**). Immunosuppressed: lifelong suppression. Refractory: suppression or TURP.

4 · CP/CPPS — aetiology

An insult (infection, stress, trauma) in a susceptible man starts **neurogenic inflammation**, then maintained by other factors.

Domain	Key evidence
Infection	Bacterial counts equal in patients and controls; mouse **CP1 strain** pain persists after clearance
Inflammation	Biopsy inflammation in ~⅓, not correlated; symptoms track **cytokines** (MIP-1α, IL-17)
Neurological	**Central sensitisation**; normal sensory thresholds; insula changes
Pelvic floor	Tenderness, **failure to relax**; tender in **81%**
Psychosocial	**Catastrophising, helplessness predict** pain; abuse history
Endocrine	Low, blunted ACTH

5 · CP/CPPS — clinical features and evaluation

- P1 P4 Chronic Prostatitis: diagnosis, classification and management; Chronic non bacterial prostatitis — *pain sites, NIH-CPSI domains, DRE/trigger points, 2-glass test, cytology, cystoscopy, UPOINT-directed multimodal therapy*
- **Pain defines it** (vs BPH): **perineum** 63%; ejaculatory pain ~half; more sites = more severe.
- **NIH-CPSI** †0–43 (pain 0–21, urinary 0–10, QOL 0–12); fall of **≥6 points** = meaningful. No bladder items → **GUPI**.
- MAPP: pain and urinary symptoms are separate factors (depression tracks pain).

- **Bladder pain in men**: ~¾ painful filling; ~42% meet IC/BPS *→ Ch 28*; **Hunner lesions** ~12% → fulguration.
- ~70% flare yearly; ED 15–40%; anxiety/depression ~2×.
- Onset clue: stress → pelvic floor; trauma → neuromuscular; post-UTI/STI → infection.
- Tenderness is **muscular, not prostatic** (47% vs 6%). DRE nodule → cancer work-up. Confirm haematuria ≥3 RBC/HPF. **PSA not indicated**; never ascribe a raised value to CP/CPPS.
- Questionnaires: NIH-CPSI/GUPI, AUA-SI (voiding), SHIM (ED). Optional imaging: CT if abdominal pain, scrotal US for testicular pain, lumbosacral MRI if radiculopathy; no routine pelvic MRI or prostate biopsy. Cystoscopy if low flow/high PVR or pain on filling relieved by voiding. Uroflow (young man, poor stream → stricture). Videourodynamics after medical failure: bladder-neck obstruction or **pseudodyssynergia** (sphincter activity during voiding *without* straining) → physiotherapy.

UPOINT phenotyping

Monotherapy fails because domains differ ("snowflake"); UPOINT maps domains for simultaneous multimodal therapy; domain count tracks severity (UK, ICUD).

Domain	Identify by	Therapy
Urinary	LUTS, PVR	α-blocker; †antimuscarinic
Psychosocial	Depression, catastrophising	Psychology, stress reduction
Organ-specific	Prostate tenderness; bladder pain	†Quercetin/pollen; bladder therapy; 5-ARI in BPH
Infection	VB3+ with VB2−	Antibiotics
Neurologic/systemic	Pain beyond pelvis, IBS, fibromyalgia	Tricyclics, pregabalin, acupuncture
Tenderness	Pelvic floor trigger points	Physiotherapy, myofascial release

6 · CP/CPPS — treatment

KEY POINT

Governing fact: no monotherapy has a positive definitive trial — each tested one agent in an unphenotyped population. Start conservative, then give **simultaneous, phenotype-directed multimodal therapy** with a multidisciplinary team.

Modality	Evidence / role
Antibiotics	Cipro/levofloxacin trials negative; benefit may be anti-inflammatory (↓ IL-6, TNF-α) or VB3-only bacteria. EAU: quinolone or tetracycline ≥ **6 weeks** only if **treatment-naive and <1 year**. **No repeated courses without positive culture**.
α-blockers	Tamsulosin, alfuzosin trials negative. Newly diagnosed, naive men with voiding symptoms / <1 year. Best combination: **antibiotic + α-blocker**.
Anti-inflammatories	Celecoxib helps pain only; steroids negative.
5-ARI	Dutasteride (REDUCE) ≥6-point NIH-CPSI fall 49% vs 37%; older men with BPH; counsel sexual side effects.
Neuropathic	Pregabalin missed endpoint; amitriptyline ≥50 mg (underdosing = false failure).
Conservative	Diet; **aerobic exercise**; cognitive therapy.

Modality	Evidence / role
Acupuncture	Strongest conservative evidence: sham-controlled RCT 61.5% vs 38.3% responders at 32 weeks. Offer after first-line failure or if drugs declined.
Pelvic floor physiotherapy	Trigger-point release + paradoxical relaxation (Stanford); myofascial therapy > global massage (57% vs 21%). Refer if tender; works in pseudodyssynergia; UDS do not predict response; use experienced therapists.
Neuromodulation	PTNS; SNS if frequency/urgency (off-label for pain).
Surgery	No TURP/prostatectomy. **Bladder neck incision** only for proven obstruction after excluding pseudodyssynergia. Hunner fulguration.

7 · Acute orchitis

- **Bacterial** (>35 y): Gram-negatives; also syphilis, TB, brucellosis, filariasis, candiduria.
- **Mumps**: commonest viral (even in vaccinated); parotitis then orchitis 2–3 weeks after exposure; **atrophy in ~half**, rarely infertility; interferon may not prevent atrophy. Also HIV, EBV, SARS-CoV-2.
- **Autoimmune**: primary (anti-sperm antibodies); secondary vasculitis (Behçet, PAN, HSP) → steroids, immunosuppressants, IVIG.
- **Pain duration** (torsion is the key differential); urine culture (~40% positive); STI tests; **scrotal US**. Treat culture/STI; HIV → suppression (*Salmonella*). **Follow-up mandatory** (missed torsion/tumour); pain settles 1–3 days, swelling 2–4 weeks.

8 · Acute epididymitis

- Epididymal pain/swelling **<6 weeks**; **ascending** (~80% urine concordance).
- **<35 y sexually active → *C. trachomatis, N. gonorrhoeae*; >35 and children → enteric**. >40% under 35 grow enterics → **STI-screen all sexually active men**.
- Pain starts in the **tail**; unilateral 96%; fever ~25%; orchitis half; abscess <10%.
- FBC, culture, first-void urine, STI NAAT; PVR in older men. **Doppler US** excludes torsion.

Likely pathogen	Regimen
Chlamydia/gonorrhoea	Ceftriaxone **500 mg IM** single (1 g if ≥150 kg) + doxycycline 100 mg bd **10 days**
STI + enteric (insertive anal sex)	Ceftriaxone 500 mg IM + levofloxacin 500 mg od 10 days
Enteric only	Levofloxacin 500 mg od 10 days

- Rest, elevation, NSAIDs; admit if febrile with leucocytosis. STI → HIV test, treat partners. **No improvement at 72 h** → repeat US (abscess, infarction → surgery).

9 · Chronic orchitis and epididymitis

- Testicular pain >3 months without infection = **chronic orchialgia / scrotal content pain**.
- **Granulomatous orchitis**: TB, brucellosis, syphilis, leprosy, sarcoidosis; **idiopathic** usually found after orchidectomy for suspected tumour (IgG4 may contribute). **Malakoplakia**: **Michaelis–Gutmann bodies**.
- **Chronic epididymitis** (≥**3 months**), **Nickel classification**: inflammatory (infective, post-infective, TB, **amiodarone**, idiopathic); obstructive (congenital, post-infective, post-

vasectomy); epididymalgia (normal epididymis). Culture, US; reassurance.

- **TB epididymitis**: commonest genital site (sole in 45%); painless swelling, **sinus**. Four drugs ≥**6 months**; surgery **only on anti-TB drugs**. **Mass not shrinking after 3 weeks → inguinal exploration**.

10 · Chronic scrotal content pain (CSCP)

Intrascrotal pain ≥**3 months** interfering with life; no evidence-based guidelines.

- Peak 20–30 y. Post-procedural: **inguinal hernia repair** (commonest; mesh/recurrent), vasectomy (~**1 in 1000**, consent figure), scrotal surgery. Others: varicocele, hydrocele, intermittent torsion, referred pain (ureteric stone, disc, aneurysm). Idiopathic 18–25%.
- **Pelvic floor dysfunction** common (93% symptom, 88% high tone). **Wallerian degeneration** (84% vs 20%) in the **"trifecta nerve complex"**: cremaster, perivasal sheath, posterior lipomatous tissue.
- **First decision**: isolated scrotal pain or part of CP/CPPS? Check **testosterone, vitamin B12**; COSI score; **duplex US**.
- **Spermatic cord block**: **20 mL 0.25% bupivacaine** at the **pubic tubercle**; **>50% relief** predicts MDSC success.
- Medical: NSAIDs, **neuropathic agents** (no chronic opioids); antibiotics rarely. Serial bupivacaine + triamcinolone blocks (~70% sustained). Physiotherapy if abnormal tone on DRE. Botox: placebo effect on RCT; pulsed radiofrequency unproven.

> **ALGORITHM**
> Identifiable lesion → directed surgery (hydrocelectomy, spermatocelectomy, orchidopexy, **microsurgical varicocelectomy**) → post-vasectomy → **vasectomy reversal** (82–93%; ≥3 months) → no lesion + positive block → **MDSC** → focal epididymal → epididymectomy → last resort: **inguinal orchidectomy**

- Varicocelectomy predictors: >7 veins, pain >6, >9 months.
- **MDSC** (Devine–Schellhammer 1978): subinguinal, microscopic; divide cord nerves, **preserve arteries and lymphatics** (prevents hydrocele), ligate internal spermatic veins; cure 50–70%, better if several structures painful, prior failed surgery no barrier; consent for **testicular atrophy**. **Targeted trifecta MDSC** (cremasteric fibres, vasal sheath, posterior fat): equal efficacy, ~21 vs 53 min.
- **Epididymectomy**: 50–75%; best with focal lesion; include vasectomy site.
- **Orchidectomy**: least reliable (0–75%); **inguinal > scrotal** (75% vs 55%); better for nociceptive pain. Persistent pain → resect genital branch of genitofemoral nerve.

11 · Landmark trials

Trial	Design	Result	Take-home
NIH CPCRN ciprofloxacin/tamsulosin	RCT, 6 weeks, NIH-CPSI ≥15	No benefit vs placebo	Monotherapy fails in unselected CP/CPPS
REDUCE (CP/CPPS analysis)	Dutasteride vs placebo, 4 years	≥6-point fall 49% vs 37%	5-ARI for older men with BPH
MAPP myofascial physiotherapy	Feasibility vs global massage	57% vs 21% responders	Physiotherapy for tender pelvic floor

Trial	Design	Result	Take-home
Acupuncture vs sham	Large RCT	61.5% vs 38.3% at 32 weeks	Evidence-based option
Botulinum toxin cord injection	Double-blind RCT	No better than local anaesthetic	Open-label series overestimate benefit

28 Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

1 · Definition and nomenclature

- P2 P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — *AUA, ESSIC (types 1–3, A–C), NIDDK, ICD-11 definitions; IC vs BPS split; phenotypes*
- **IC/BPS** — heterogeneous chronic pelvic pain syndrome: from inflammatory bladder disease with **Hunner lesions (HLs)** to bladder pain with many non-urological complaints and a normal bladder.
- **AUA**: bladder-related pain, pressure or discomfort with LUTS **>6 weeks**, no infection/other cause. † **ESSIC (2008)**: **>6 months** + ≥1 urinary symptom; **type 3C** = classic HL disease (C = inflammatory infiltrate/detrusor mastocytosis/granulation tissue/fibrosis).
- **NIDDK (1987)**: research definition; clinically missed **~60%** → definitions broadened.
- **ICD-11** "chronic primary bladder pain syndrome": classes the pain itself as the disease (pain worse on filling, frequency).
- **UCPPS** (MAPP Research Network) = IC/BPS (both sexes) + CP/CPPS (men) → *Ch 27*.
- **Trend**: "IC" = HL disease; "BPS" = non-HL. BPS never shown to progress to HL → probably distinct disorders.

2 · Epidemiology (headline)

- RICE study: **3.3–7.9 million US women**, 2.1–4.6 million men with IC/BPS-type symptoms. Mean onset 30–40 years.
- F:M **5:1** in clinics but **2–3:1** in populations → men under-diagnosed. **Ask men labelled CP/CPPS about bladder pain**; reconsider IC/BPS when CP/CPPS treatment fails.
- **HL prevalence ~20%** (3.5–56%); rises with age — uncommon <30, >half over 60–70. Same in men.
- **Natural history**: chronic, **not progressive**; flares (commonest trigger sexual activity; diet, abdominal exercise). ~⅓ remit ≥3 months.
- Childhood adversity is a risk factor; catastrophising/depression predict suicidal ideation → CBT.

> **KEY POINT**
> Uncontrolled series "work" but RCTs fail: regression to the mean, placebo, extra attention. Expect improvement in the first 3–4 weeks of any treatment.

3 · Aetiology and pathophysiology

Unknown, multifactorial; mechanisms combine into a self-perpetuating cycle.

Theory	Mechanism / evidence	Treatment link
GAG layer / "leaky" urothelium	Urinary solutes (potassium, "toxic cations") reach suburothelial nerves → pain, inflammation	PPS, heparin, hyaluronic acid, chondroitin; potassium test
Mast cells	Histamine release → pain, fibrosis. Marked in HL disease, sparse in BPS. Counts have **no diagnostic role**	Hydroxyzine, cimetidine, amitriptyline

Theory	Mechanism / evidence	Treatment link
Neurogenic	Neurogenic inflammation, pelvic organ cross-sensitisation, central sensitisation, HPA-axis dysregulation; small-fibre polyneuropathy in some	Tricyclics, gabapentinoids, neuromodulation, CBT
Autoimmune	Large relative excess of SLE, Sjögren, RA, IBD; lymphoplasmacytic infiltrate in HL	CyA for HL disease
Infection	No consistent organism, even by microbiome. 18-week antibiotic RCT: no benefit, more adverse events	No antibiotics without proven infection

KEY POINT

Central sensitisation explains why many BPS patients have no physical findings and do not respond to local bladder therapy — the mechanism is systemic, not vesical.

4 · Pathology: Hunner lesion vs non-Hunner phenotype

- **No pathognomonic histology**; biopsy normal in ~half. Biopsy only HL/suspicious lesions — exclude CIS, cancer, eosinophilic, TB, ketamine cystitis.

Feature	HL disease ("IC")	Non-HL ("BPS")
Cystoscopy	Stellate erythema, radiating vessels, central scar; bleeds on filling ("waterfall"); rare at trigone	Normal, or glomerulations after distension
Histology	Ulceration, granulation, dense **lymphoplasmacytic** infiltrate, ↑mast cells; denudation beyond lesion	Near-normal; sparse inflammation
Pain source	Bladder — end-organ disease	Often regional/widespread
Treatment	Fulguration / triamcinolone; CyA; DMSO better; cystectomy succeeds	Phenotype-directed; surgery often fails

5 · Associated conditions

- **75–80%** have pain beyond the pelvis; ~38% widespread. True somatisation rare.
- **COPCs**: **IBS** commonest (>30%); **fibromyalgia** (prototype nociplastic pain); CFS, migraine, TMJD, low back pain.
- **Regional**: **HTPFD** in **~80%**; vulvodynia; endometriosis.
- **Autoimmune** — largest relative excess: IBD (~100-fold), SLE (tens-fold), SS and RA (10–15-fold). **Trap**: in SLE think **lupus cystitis** (fibrotic bladder + hydroureteronephrosis).

KEY POINT

Commonest ≠ most over-represented: IBS is commonest; autoimmune disease shows the greatest relative excess. Widespread pain predicts failure of bladder-directed therapy.

6 · Clinical features

- Suprapubic pain **worse with filling, relieved by voiding** → day/night frequency. Dyspareunia common; isolated urethral pain uncommon.

Feature	OAB	IC/BPS	HTPFD
Urge	Sudden; fear of leakage	Persistent, mounting, pain-driven	Persistent; pelvic pressure
Pain	Absent	Present	Present
Incontinence	UUI	Not a feature	—

Feature	OAB	IC/BPS	HTPFD
Voiding	—	Frequency, low volumes	Obstructive

7 · Diagnosis — exclusion plus localisation

Clinical diagnosis of exclusion; missed pain generators are a common cause of treatment failure.

Basic evaluation (AUA 2022)

- **History**: pain–filling link, triggers, sexual pain, bowel, pelvic surgery/irradiation, **cyclophosphamide, ketamine**.
- **Examination**: women — diverticulum, atrophy, lichen sclerosus, mesh, **Q-tip test** (vulvodynia); men — genitals, DRE; **pelvic-floor palpation**; neurology.
- **Tests**: PVR, urinalysis, culture; cytology if risk (smoking, haematuria, age >40). 24-h frequency–volume chart — large voids → non-bladder source.
- **Scores**: O'Leary–Sant ICSI/ICPI (trial endpoint ≥30% fall), GUPI, BPIC-SS, VAS, GRA. † PUF scale screens, not diagnostic. Score pain and urinary symptoms separately.

Confusable disorders (ESSIC-modified)

Group	Disorder	Excluded by
Malignancy	Bladder cancer/CIS; prostate, gynae, colon	Cystoscopy ± biopsy; PSA; pelvic exam
Infection	Bacterial, atypical, **TB**, herpes; prostatitis	Culture; **sterile pyuria → TB culture**
Iatrogenic/drug	Radiation, cyclophosphamide, tiaprofenic acid, **ketamine**	History; ketamine — upper-tract change
Outflow	BNO, BPO, retention	PVR, uroflow, (video)UDS
Structural/other	Stones, diverticulum; endometriosis, vulvodynia, atrophy; IBS, IBD	Imaging, cystoscopy, exam, laparoscopy, colonoscopy
Neuromuscular	Pudendal entrapment; pelvic-floor myalgia	Nerve block; pelvic-floor exam

Ketamine cystitis

- NMDA antagonist; HL-like inflammation → fibrosis, **hydronephrosis (~30%)**, renal insufficiency (~10%). Suspect in the **young (<30) with visible inflammation**. Treatment: **cessation**.

High-tone pelvic floor dysfunction

- Pelvic pressure, persistent urge, obstructive voiding, constipation, next-day dyspareunia. Tender hypertonic muscle, **myofascial trigger points**.
- Six-site exam: R/L anterior levator, posterior levator, obturator internus (vaginal/transrectal). Missed HTPFD → poor response → pelvic-floor physiotherapist.

Urodynamics

- Not standard (AUA); for failed therapy, incontinence, high PVR, obstruction. Reduced capacity, hypersensitivity, **normal compliance**; fibrotic end-stage → low compliance. DO in 15–19% — **does not exclude IC/BPS**.

Office cystoscopy

- **AUA 2022**: when **HL suspected, especially >50**; also haematuria, failed therapy, suspected stone/cancer.

- Flexible cystoscopy shows most HLs **without hydrodistension**. Biopsy CIS-like and **any trigonal lesion**. NBI may help.

Anaesthetic challenge vs potassium test

- **Anaesthetic challenge** (15 mL 0.5% bupivacaine + 15 mL 2% lidocaine): relief → bladder-centric, selects intravesical therapy; no relief does not exclude.
- **PST**: **0.4 M KCl** vs water. **Not recommended** — painful; negative in ¼ of NIDDK patients, positive in ~⅓ normal men, ¼ OAB, most CP/CPPS. Does not predict PPS response.

Cystoscopy with hydrodistension (CHD)

- Under anaesthesia **80 cm H_2O for 1–3 min** → drain → refill → **glomerulations**; tears near bladder neck.
- **Glomerulations non-specific** (radiotherapy, carcinoma, chemotherapy, dialysis/diverted bladders, normal overdistended bladders); don't correlate with severity.
- Biopsy only **after** distension (perforation risk), for suspicious lesions — CIS has been misdiagnosed as IC/BPS.

8 · Phenotype-directed management

- P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — *four-phenotype table: HL, anaesthetic-responsive, pelvic floor, widespread → matched treatment*
- Pain centre: HL and anaesthetic-responsive = bladder; HTPFD = pelvic floor; widespread = central (multidisciplinary care).

> **KEY POINT**
> MAPP reanalysis (2023) of two negative NIH RCTs: hydroxyzine/PPS helped **localised pelvic pain**; amitriptyline helped **widespread pain**. A trial negative in all-comers can hide a positive result in the right phenotype.

9 · Treatment — AUA 2022 principles

- **No "home run"**; goal is QoL and coping. Set expectations; agree a flare plan.
- **No tiers**: grouped options; may move to any category **except major surgery** without failing "lower" ones; severe symptoms may justify initial multimodal therapy.
- Stop ineffective treatment; not improving → look for other pain generators.
- **Standard**: manual (myofascial) PT for pelvic-floor tenderness. **Recommendation**: fulguration and/or triamcinolone for HLs. Most drugs are **Options**.
- CBT; pudendal block; trigger-point injections (anaesthetic or BTX-A). Chronic opioids not routinely recommended.

> **ALGORITHM**
> Symptoms ≥6 weeks → history, exam (pelvic floor, Q-tip, DRE), PVR, urinalysis, culture ± cytology → exclude confusable disease → cystoscopy if HL suspected (>50) → HL: fulguration/triamcinolone → refractory: CyA
> No HL → phenotype: pelvic floor → physiotherapy; lidocaine responder → intravesical; widespread → systemic/CBT → behavioural + oral + intravesical → BTX-A or SNM → (rare, bladder-centric, all failed) diversion ± cystectomy

10 · Treatment options (AUA 2022)

- P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — *options table: behavioural, physical (avoid Kegel), oral, intravesical, procedures, refractory HL, last resort*

10.1 Behavioural and pelvic-floor therapy

- **Diet triggers**: caffeine, alcohol, carbonated drinks, artificial sweeteners, spicy food, citrus/cranberry, tomato. **Elimination diet**: diary → bland diet → reintroduce one at a time → avoid confirmed triggers only.
- **EBMP** (NIH amitriptyline RCT control arm): 45% improved alone.
- Myofascial PT RCT: 10 sessions vs global massage → **59% vs 26%**. Adjuncts: reverse Kegels, heat, relaxants, constipation control, trigger-point injections.

10.2 Oral therapy

- **Amitriptyline** (anticholinergic, SNRI-like, potent H1, sedative): **10 mg nocte, +10 mg weekly to ~50 mg** (25–75). NIH RCT 55% vs 45% (NS); **77% vs 53%** if ≥50 mg tolerated; independent of HL. **CI**: long QT, conduction block, recent MI, heart failure, ventricular arrhythmia; caution with incomplete emptying.
- **Hydroxyzine** (H1): 25 mg nocte → 50 mg nocte + 25 mg mane; RCT negative; allergy, sleep.
- **Cimetidine** (H2) 400 mg twice daily; one positive RCT.
- **PPS 100 mg three times daily**; only **3–6%** reaches urine; acts in **3–6 months**; only FDA-approved oral IC drug. Phase IV RCT negative (≈40% both arms); less effective in HL. Adverse (<4%): alopecia, GI upset, rash, bleeding; caution in high breast-cancer risk.
- **PPS maculopathy**: cumulative-dose, may be irreversible/progress after stopping. **FDA (2020)**: ophthalmic history in all; baseline exam if eye disease; retinal exam for all **within 6 months**, then periodically.
- **Ciclosporin A**: HL refractory to fulguration/triamcinolone; beat PPS in open RCT; few non-HL respond. **1–3 mg/kg/day in two doses**; judge at 3–4 months. Toxicity 30–55%: **nephrotoxicity, hypertension**, infection, malignancy, gum hyperplasia. Monitor BP, renal/liver function, level (1 month, then 3-monthly).
- Gabapentinoids (no RCTs); sildenafil, quercetin (small studies).
- **Antibiotics**: AUA — **no long-term antibiotics**; one empirical course if never treated; prophylaxis only for proven recurrent cystitis triggering flares.

10.3 Intravesical therapy

For **bladder-centric** pain (lidocaine responders); poor lidocaine response predicts failure of every instillation. AUA Option.

Agent	Regimen	Key points
DMSO 50%	50 mL, **15–20 min dwell**, every 1–2 weeks × 4–6	FDA-approved stand-alone. Anti-inflammatory, analgesic, dissolves collagen. Better in HL; DO patients don't respond. Garlic breath; initial flare; longer dwell → pain. Boosts co-instilled lidocaine absorption → toxicity.
Heparin	10,000–40,000 U, 30–60 min, self-instilled	GAG mimic; benefit takes months; better with alkalinised lidocaine.
Alkalinised lidocaine	200 mg + 8.4% bicarbonate daily × 5	RCT 30% vs ~10% at day 3. Immediate → flares. Tachyphylaxis.

Agent	Regimen	Key points
Hyaluronic acid, chondroitin, PPS	Weekly then monthly	Weak/negative trials

- Cocktails (DMSO + triamcinolone + bicarbonate + heparin ± lidocaine) weekly × 6 — none proved better than DMSO alone. Watch for lidocaine toxicity (perioral numbness) → drain.

10.4 Hunner lesion treatment

- **AUA Recommendation**: fulguration and/or submucosal triamcinolone; biopsy if never biopsied. **75–90%** improve; repeat at mean 7–12 months (triamcinolone) or 11–29 months (fulguration).
- Resection risks perforation; laser risks bowel injury (forward scatter). Treating glomerulations does not help.

ALGORITHM

GA → map HLs before distension bleeding → mild distension → biopsy suspicious/trigonal lesions → fulgurate at **low volume** (Bugbee) from the **outer margin inwards** → triamcinolone 40 mg/mL (or 1:1 with bupivacaine/lidocaine), 0.5-mL aliquots, 2–3 mm submucosal → fulgurate bleeders → empty, instil lidocaine–bupivacaine

KEY POINT

Fulgurate from the margin inward: cautery erythema/oedema spreads peripherally, so the true edge is lost if you start centrally.

10.5 Hydrodistension

- AUA Option: **short (1–3 min), low-pressure (60–80 cm H_2O)**; instil anaesthetic at end. Benefit: ~½ at 1 month, ¼ at 3, ~none at 6. Capacity <600 mL responds better; post-procedure flare common.
- Anaesthetic capacity **<400 mL** = bladder-centric phenotype (often HL).

10.6 OnabotulinumtoxinA and neuromodulation

- **BTX-A**: Option after failed conservative/oral/intravesical; not FDA-approved for IC/BPS; acts on sensory pathways. **100 U** in 5 mL (200 U → dysuria/retention), into **trigone** (trigone-including vs sparing equal). Lasts ~5–9 months; repeatable. **Must accept CIC**.
- TENS adjunct only; PTNS sham-negative. **SNM (S3)**: Option; best with **HTPFD/levator tenderness**; IC/BPS has the highest SNM adverse-event/explantation rates.

	BTX-A	SNM
Favoured when	Bladder-centric pain; can do CIC	Obstructive voiding, incomplete emptying, HTPFD, CIC impossible
Main risk	Retention → CIC	Adverse events, explantation

10.7 Major surgery — last resort

- **Rare (<10%), selected, bladder-centric** patients failing all else; pain may persist after cystectomy.
- Work-up: body map → pelvic-floor exam → intravesical anaesthetic trial → treat other generators. **Differential spinal block**: pain persisting after spinal to T10 → surgery will fail.

Predicts success	Predicts failure
HLs (refractory)	No HLs

Predicts success	Predicts failure
Small fibrotic bladder; anaesthetic capacity <200–400 mL	High anaesthetic capacity
Profound relief with intravesical anaesthetic	Equivocal/no response
Painful filling, pain on scope touch	Urethral/neuropathic pain; HTPFD; widespread pain

- Conduit with bladder in situ (~⅕ later cystectomy); cystectomy + conduit (~¾ pain-free); continent pouch (pouch pain 40–50%); supratrigonal augmentation (trigonal cuff recurrence, CIC). Prefer the **simplest diversion**; retained bladder risks pain, pyocystis, haemorrhage.

11 · What NOT to do (AUA)

Do not offer	Why
Long-term oral antibiotics	RCT negative; adverse events 80% vs 40%
Intravesical BCG	Ineffective; DMSO superior; † BCG sepsis
High-pressure, long hydrodistension †	No benefit; rupture, sepsis †
Long-term oral glucocorticoids	No benefit; toxicity
† Intravesical resiniferatoxin	Ineffective in RCT †
Kegel strengthening	Worsens HTPFD pain
Major surgery first / non-bladder-centric pain	Irreversible; pain persists

KEY POINT

Nothing is worse than irreversible surgery followed by continued pain — every predictor of surgical success marks bladder-generated pain.

12 · Landmark trials

Trial	Design	Result	Take-home
Myofascial PT RCT (FitzGerald 2012)	10 sessions myofascial vs global massage	59% vs 26%	Guideline Standard
NIH amitriptyline RCT (Foster 2010)	EBMP ± amitriptyline	55% vs 45% (NS); 77% vs 53% if ≥50 mg	Titrate to ≥50 mg; behavioural therapy works
PPS phase IV RCT	PPS vs placebo	No difference (≈40% each)	Efficacy doubtful; remains Option
MAPP phenotype reanalysis (2023)	Two negative NIH RCTs by pain distribution	Hydroxyzine/PPS localised; amitriptyline widespread	Treat by phenotype

29 Sexually Transmitted Diseases

1 · Screening, prevention and partner notification

- Chlamydia = commonest reportable STD; syphilis = fastest rising.
- **NAAT** is the standard screen; MSM need urethral, rectal and pharyngeal tests. MCQ Test the pharynx for gonorrhoea, not chlamydia.
- **Condoms** cut HIV acquisition ~80%; protect only covered skin; lambskin lets viruses through.
- **Partner notification**: previous **60 days** for urethritis, chlamydia, *M. genitalium*; all partners for trichomoniasis; previous month for scabies/lice. Abstain 7 days and until partners treated.
- Test everyone with/at risk of an STD for HIV and syphilis.

2 · Urethritis

- Discharge, dysuria or none. Complications: epididymitis, orchitis, prostatitis.
- **Gonococcal**: 5–20%. **NGU**: *C. trachomatis* > *M. genitalium* (leading cause of **persistent/recurrent** urethritis); less often *T. vaginalis*, HSV, adenovirus; no pathogen in up to half.
- **Diagnosis** — any one: discharge; Gram stain **≥2 white blood cells (WBC) per oil-immersion field** (old ≥5 missed many); positive leucocyte esterase on first-void urine; ≥10 WBC/hpf in first-void urine.
- Intracellular gram-negative diplococci in a symptomatic man confirm gonorrhoea; negative Gram stain does not exclude.
- **First-void urine NAAT** preferred (more sensitive). **Culture** for treatment failure, resistance surveillance, children — take it before other swabs.

Pathogen	First line	Alternatives / notes	Follow-up
Gonorrhoea (CDC 2021)	**Ceftriaxone 500 mg IM once** (1 g if ≥150 kg)	Gentamicin 240 mg IM + azithromycin 2 g (β-lactam allergy); cefixime 800 mg. Europe: ceftriaxone 1 g + azithromycin 2 g. Add chlamydia cover unless excluded	Test of cure only for **pharyngeal**; retest at 3 months
Chlamydia / NGU	**Doxycycline 100 mg bd 7 days**	Azithromycin 1 g once	No test of cure; retest 3 months
M. genitalium	Doxycycline 7 days, then **azithromycin** 1 g + 500 mg daily × 3 if macrolide-sensitive, or **moxifloxacin** 400 mg × 7 days if resistant/untested	Single-dose azithromycin inadequate	Abstain until cured
T. vaginalis (male)	Metronidazole 2 g once	Tinidazole 2 g once	Treat all partners

- **Ceftriaxone monotherapy**: rising resistance → higher dose, azithromycin dropped. Cephalosporins safe unless IgE-mediated penicillin reaction in last 10 years (cross-reactivity <1%).

- **Gonorrhoea treatment failure**: symptoms >3–5 days or positive test of cure without re-exposure → culture + susceptibility.
- ***M. genitalium***: no cell wall (Gram-invisible), culture takes months → NAAT + macrolide-resistance assay. Doxycycline lowers load; resistance result picks the second drug.
- **Trichomonas** not in first NGU work-up: test partners of infected women, high-prevalence areas, empiric failure.
- No chlamydia test of cure — nucleic acid persists ~3 weeks.

ALGORITHM

Persistent NGU → check adherence and untreated partner → NAAT for *M. genitalium* ± *T. vaginalis* → negative → think chronic pelvic pain syndrome

KEY POINT

Men with gonorrhoea usually have symptoms; women are usually silent and present with PID or infertility. Treating the man protects his partner. Chlamydia can also cause reactive arthritis.

3 · Epididymitis by age

Acute <6 weeks; chronic (≥6 weeks) mostly TB → *Ch 27*. Always exclude **torsion** (US only if suspected). Urethritis criteria + NAAT; urine culture if enteric likely.

Group	Organisms	Empiric regimen (10 days)
Young, sexually active	*C. trachomatis*, *N. gonorrhoeae*	Ceftriaxone 500 mg IM + doxycycline 100 mg bd
Insertive anal sex	Above + enterics	Ceftriaxone + levofloxacin 500 mg daily
>35 years, obstruction or instrumentation	*E. coli*	Levofloxacin alone

10 days, not 7. Doxycycline for chlamydia, ceftriaxone for gonorrhoea, levofloxacin for enterics. Abstain until patient and partners treated; test for STDs/HIV.

4 · Genital ulcer disease

Herpes commonest, then syphilis. Appearance misleads → **every ulcer**: syphilis serology ± darkfield/PCR; HSV NAAT + type-specific serology; *H. ducreyi* test if endemic; HIV. Biopsy atypical/non-healing ulcers (carcinoma mimics).

Disease (organism)	Lesion	Nodes	Test	Treatment
Primary syphilis (*T. pallidum*)	**Painless**, indurated, clean, single	Painless, rubbery, bilateral	Serology; darkfield	Benzathine penicillin G 2.4 MU IM once
Herpes (HSV-2 > HSV-1)	**Painful** multiple vesicles → shallow ulcers	Tender, bilateral	NAAT of unroofed base	Aciclovir and related drugs
Chancroid (*H. ducreyi*)	**Painful, ragged, undermined, purulent**	Tender **suppurative buboes**	Special culture; exclusion	Azithromycin 1 g once, or ceftriaxone 250 mg IM once, or ciprofloxacin 500 mg bd 3 days, or erythromycin 7 days

Disease (organism)	Lesion	Nodes	Test	Treatment
LGV (*C. trachomatis* L1–L3)	Small, painless, transient	Painful, matted, fistulae; †groove sign	Chlamydia NAAT	Doxycycline bd **21 days**
Donovanosis (*Klebsiella granulomatis*)	**Painless, beefy red**, bleeds easily	**Absent** (†pseudobuboes)	**Donovan bodies** on crush preparation	Azithromycin 1 g weekly or 500 mg daily, ≥3 weeks **and until healed**

Genital herpes

- Latent in **sacral ganglia**; reactivated by trauma (incl. surgery), UV, immunosuppression, fever.
- Asymptomatic shedding ~10% of days → most transmission from unaware carriers. HSV-2 recurs more → type it. Negative NAAT does not exclude; Tzanck insensitive.
- Complications: aseptic meningitis, **urinary retention** (autonomic). HSV-2 raises HIV risk 2–3-fold.

Drug	First episode (7–10 days)	Suppressive	Episodic
Aciclovir	400 mg tds	400 mg bd	800 mg bd 5 days
Valaciclovir	1 g bd	500 mg–1 g daily	500 mg bd 3 days
Famciclovir	250 mg tds	250 mg bd	1 g bd 1 day

- Treat every first episode. **Suppressive** also cuts transmission. **Episodic** must start within 1 day of lesions or in prodrome.

Chancroid, donovanosis, LGV

- **Chancroid** needs all four: painful ulcer, typical picture, syphilis excluded, HSV excluded. Aspirate/drain fluctuant buboes. HIV and uncircumcised men respond less well.
- **Donovanosis**: antibiotic failure → biopsy (**penile SCC mimics it**).
- P1 Lymphogranuloma venereum — *L1–L3; proctocolitis in MSM now commonest; groove sign; elephantiasis, rectal stricture; inguinal disease commoner in men; doxycycline 21 days*

5 · Syphilis staging and serology

Stage	Features	Treatment (penicillin-allergic)
Primary (10–90 days)	Chancre + nodes; heals in 3–8 weeks	Benzathine penicillin **2.4 MU IM once** (doxycycline 14 days)
Secondary (3–5 months)	Rash on **palms and soles**, **condylomata lata**, alopecia, hepatitis, nephritis	
Early latent (<1 year)	Positive serology only	
Late latent / unknown	Positive serology only	Benzathine **weekly × 3** (doxycycline 28 days)
Tertiary	Gumma, aortitis, tabes dorsalis	Weekly × 3 if CSF normal
Neurosyphilis (any stage)	CNS, eye, ear	**Aqueous penicillin G 18–24 MU/day IV 10–14 days**

- Cannot be cultured → **two serological tests**:
  - **Nontreponemal (NTT)** — VDRL, RPR: false positives (HIV, autoimmune, pregnancy); titre falls with treatment → **monitors treatment**.
  - **Treponemal (TT)** — FTA-ABS, TP-PA, EIA: usually positive for life.
- Significant change = **fourfold (two dilutions)**; use the same NTT each time.

> **ALGORITHM**
> **Reverse algorithm**: TT positive → NTT → NTT negative → second, different TT → positive without past treatment = treat as late latent

- **Jarisch–Herxheimer**: fever, myalgia within 24 h — treponemal lysis, not allergy; NSAIDs.
- Penicillin allergy + doubtful adherence → desensitise, give benzathine (IM only; neurosyphilis needs IV aqueous).
- Reverse-algorithm TT positive with documented past treatment: no action unless re-exposure (repeat NTT in 2–4 weeks).
- Primary/secondary: test HIV, offer PrEP if negative.
- **Follow-up** serology at 6 and 12 months (+24 for latent). Fourfold rise = failure/reinfection → retreat weekly × 3, retest HIV, consider CSF. In HIV serology may be atypical → biopsy/darkfield.

6 · HPV and anogenital warts

- HPV **6 and 11**: ~90% of warts; **16 and 18**: >70% of cervical cancer, plus penile and anal.
- **Bowenoid papulosis** (HPV-16) = HSIL. **Giant condyloma** (†Buschke–Löwenstein): verrucous carcinoma from HPV-6; invades deeply despite bland histology.
- Clinical diagnosis; no typing. Inspect anal canal and meatus. **Biopsy if**: diagnosis uncertain, no response/worsening, immunocompromised, pigmented/indurated/fixed/ulcerated.

Treatment	Key points
Imiquimod 5% (patient)	3 nights/week up to 16 weeks; wash off after 6–10 h. Immune stimulation. Weakens condoms
Podofilox 0.5% (patient)	Antimitotic; bd 3 days, 4 days off. **Contraindicated in pregnancy**
Sinecatechins 15% (patient)	Green tea; tds, not washed off. Weakens condoms. Not in HIV, herpes, pregnancy
Cryotherapy, TCA 80–90% (provider)	Cryo every 1–2 weeks; TCA weekly — let dry, neutralise with bicarbonate if painful
Excision, curettage, CO_2 laser	Large warts; smoke extraction

Change treatment if no response by 3 months. Observation acceptable; treatment does not clear HPV.

- **Vaccine**: **nonavalent** (6, 11, 16, 18, 31, 33, 45, 52, 58); **11–12 years** (from 9), catch-up to **26**, both sexes. Start <15: **2 doses**; ≥15 or immunocompromised: **3 doses**. Prophylactic only; not in pregnancy or yeast allergy.

7 · Molluscum and ectoparasites

- **Molluscum**: **umbilicated** papules; regress in 6–12 months. Cryotherapy, curettage, imiquimod. In HIV extensive (≥100), worse as CD4 falls → HAART first. Biopsy atypical

lesions.

- **Pubic lice**: **permethrin 1%** rinse 10 min; resistance → malathion 0.5% or ivermectin 250 µg/kg repeated at 7–14 days. Needs live lice/viable eggs. Condoms don't protect. Review at 1 week.
- **Scabies**: nocturnal itch, burrows; delayed hypersensitivity (3–6 weeks first time, 1–3 days re-infestation). **Permethrin 5%** all skin, or ivermectin 200 µg/kg repeated at 2 weeks. **Crusted** (immunocompromised): topical + repeated ivermectin, never lindane. Lindane 1% last resort (≥10 years). Scratching → strep → post-streptococcal GN.
- Treat contacts from past month; wash bedding at 50 °C.

8 · HIV for urologists

- P1 Describe clinical implication of AIDS on the Genito-Urinary Tract? — *staging (AIDS = CD4 <200), diagnosis, GU manifestations by site (UTI, prostate, testis, voiding, ED, HIVAN, indinavir stones, cancers), PEP 72 h/28 days, needlestick 0.3%, circumcision 50–60%*
- Screen everyone aged 15–65 once.
- PEP (start within **72 h**) only for mucosal/broken-skin/percutaneous contact with blood or genital fluid — not after 72 h or for urine, saliva, tears. Needlestick (**0.3%**): three-drug PEP even if source viral load low.
- **PrEP**: daily emtricitabine + TDF, or TAF (men only).
- Circumcision (**50–60%**): inner foreskin rich in Langerhans cells; does not protect partners.
- Extras not in paper: renal TB — AFB culture of 3–6 early-morning urines, annual urine AFB for 10 years; testis reservoir → maintenance suppression (esp. *Salmonella*); indinavir stones → full metabolic work-up.

After sexual assault (men): ceftriaxone 500 mg once + doxycycline 7 days (+ metronidazole for women); HBV vaccine ± HBIG if assailant HBsAg-positive; HPV vaccine if 9–26; nPEP within 72 h.

30 Cutaneous Diseases of the External Genitalia

1 · Approach to genital dermatoses

- **History + examination**; limited morphological repertoire → mimics abound → **low threshold for biopsy**.
- Ask about OTC products (lubricants, spermicides, anti-itch creams). **Full skin survey** — extragenital lesions often give the diagnosis.
- Identify the **primary lesion**; on mucosal skin, papules/macules may present as erosions.

Primary lesion	Definition	Secondary lesion	Definition
Macule	Flat discolouration	Scale / crust	Shed keratin / dried serum
Papule	Solid, raised, **≤0.5 cm**	Erosion	Epidermis only → **no scar**
Nodule / plaque	Solid >0.5 cm (plaque superficial)	Ulcer	Epidermis + dermis → **scars**
Vesicle / bulla	Free fluid ≤ / >0.5 cm	Fissure	Linear dermal loss
Pustule	Pus	Atrophy / scar	Thinning / dermal damage
Wheal	Oedematous dermis; transient	Dermoepidermal junction decides scarring	

Bedside test	Use
KOH prep; PAS	Dermatophyte (branching hyphae), *Candida* (yeast + pseudohyphae)
Mineral oil prep	Scabies mites, eggs
Tzanck smear	HSV, varicella zoster, molluscum
Wood's lamp	Erythrasma (coral-red); vitiligo (accentuated)
Serology / culture	Syphilis / *Pseudomonas*

Biopsy: 2–3 mm punch often diagnostic; office under LA for small shaft/scrotal lesions, theatre for large/meatal.

2 · Principles of therapy

- **Topical first**; urologists overprescribe systemic antibiotics (resistance, interactions, flora disruption).
- **Systemic antifungals** only for: very extensive dermatophytosis, endemic mycoses, Majocchi granuloma, severe immunocompromise, topical failure. Avoid systemic **ketoconazole** (hepatotoxicity, adrenal insufficiency).
- **Systemic steroid**: prednisone, ≤3 weeks, single morning dose (limits HPA suppression). Long-term: osteoporosis, AVN femoral head, cataract, diabetes, hypertension.
- **Topical steroids**: thin genital dermis → atrophy, striae, telangiectasia. **Prepuce = occlusive dressing** (atrophy within 8 weeks). Rule: low potency, short course (exception: LS).
- Ointment emollients (creams sting). Topical JAK inhibitor **ruxolitinib** avoids steroid atrophy.

Physical modality	Indication	Caveat
Narrow-band UVB; 308-nm excimer	Psoriasis, vitiligo, dermatitis	Excimer: localised, no NMSC risk
PUVA	Psoriasis, vitiligo, LP	**Genital SCC** risk → shield; supplanted

Physical modality	Indication	Caveat
Photodynamic therapy	Large/resistant condyloma, warts in pregnancy	Off-label
CO_2 laser; cryosurgery	Condyloma, pearly penile papules; molluscum	Cryo more used

3 · Eczematous dermatoses

Pruritic red genital eruption: exclude *Candida*, herpes, impetigo, drug reaction first.

Atopic dermatitis (AD)

- Relapsing, flexural, intensely pruritic, ill-defined border. **90%** present <5 years; asthma/rhinitis.
- **Filaggrin defect** → leaky barrier; scratching → *S. aureus* superantigens. Spongiotic biopsy; atopic history is the clue.
- Soap substitutes, emollients, short topical steroid, calcineurin inhibitors (tacrolimus, pimecrolimus), ruxolitinib, sedating antihistamine. Oral antistaphylococcals no benefit (RCT). Severe: **dupilumab**, tralokinumab (anti-IL-13).

Contact dermatitis

Feature	Irritant (ICD)	Allergic (ACD)
Mechanism	Direct cytotoxicity; anyone; dose-driven	Type IV hypersensitivity in sensitised
Share	~**80%**	Minority; genital rare (2.4%)
Agents	Soaps, solvents, acids/alkalis	**Nickel** (commonest; piercing), neomycin, bacitracin, fragrances, preservatives, latex, topical steroids
Picture	Burning, limited to exposure zone; minutes–hours	Demarcated pruritic, weeping
Confirm	Exposure history	**Patch testing**
Treat	Avoidance, barrier	Remove allergen, antihistamine; severe → **3-week** prednisone taper (short course rebounds)

Seborrhoeic dermatitis (SD)

- Pink-yellow greasy-scaled plaques in sebaceous areas (incl. glans, pubis); uncircumcised. *Malassezia*. Up to **83%** of AIDS patients; Parkinson → **extensive SD → test for HIV**.
- Topical **antifungal** (ketoconazole) is the backbone; hydrocortisone 2.5% briefly only.

Lichen simplex chronicus (lichenification; LS differential): † break itch–scratch cycle (short potent steroid, night sedation).

4 · Papulosquamous disorders

Differential: tinea, erythrasma, secondary syphilis, neoplasia (Bowen, EPD, mycosis fungoides) — "psoriasis" failing treatment needs KOH, syphilis serology, biopsy.

Psoriasis

- ~2%; ⅓ familial; triggers: trauma (Koebner), streptococci, stress, drugs. Onset 20–30 and 50–60.

- Sharp plaque, silvery scale; check elbows, knees, nails, scalp, **umbilicus**, natal cleft. **Inverse psoriasis** loses scale in folds → KOH to exclude tinea/*Candida*.
- Histology: **"sandwich sign"** (alternating neutrophils and parakeratosis).

Step	Genital psoriasis treatment
First-line	Low-potency steroid (hydrocortisone butyrate 0.1%) **≤2 weeks** continuous; triamcinolone 0.1% for thick plaque
Steroid-sparing	Calcipotriene, calcineurin inhibitors, low-potency retinoids
Localised	308-nm excimer laser
Contraindicated	**PUVA** (dose-dependent genital SCC)
Extensive	Methotrexate, cyclosporine, retinoids, apremilast, anti-TNF, ustekinumab, **ixekizumab (IL-17) — best genital data**

Reactive arthritis (formerly Reiter)

- After urethritis (*Chlamydia*, gonococcus, *Ureaplasma*) or enteric infection in **HLA-B27**; conjunctivitis, arthritis, oral ulcers (⅓ complete).
- **Balanitis circinata** and **keratoderma blennorrhagicum** look/stain like psoriasis — diagnose from **context**. Self-limiting; low-potency steroid.

Lichen planus (LP) and lichen nitidus

- LP: **polygonal, violaceous, flat-topped papules**; glans papules, white reticular/annular; Koebner. Search for drug/contact triggers.
- Papular LP resolves in ⅔ by 1 year; **erosive mucosal LP persists, scars, SCC risk** → regular checks; biopsy ulcerated lesions.
- **Ultrapotent** steroid (clobetasol 0.05%) ± calcineurin inhibitor; severe → systemic steroid.
- **Lichen nitidus**: tiny clustered flesh-coloured papules; resolves within a year.

5 · Lichen sclerosus (LS) / BXO

- P1 Lichen sclerosus (BXO, balanitis xerotica obliterans) — *scarring disease; phimosis → meatus → anterior stricture; histology; HPV-independent SCC 2.3–9%; clobetasol 3 months → maintenance; circumcision; BMG not genital skin*
- Female 6–10×; males usually >60; **15–20%** extragenital. Male: **perianal spared**; purpura may mask whiteness.
- Differential includes erosive LP, lichen simplex, morphoea, pemphigoid, child abuse.

> **ALGORITHM**
> Biopsy → **potent steroid (clobetasol 0.05%) for 3 months** → taper → twice-weekly maintenance (or mometasone 0.1% / calcineurin inhibitor) → lifelong surveillance of non-healing fissures, ulcers, nodules

- Recurrence **40–50%** → proactive maintenance. Calcineurin inhibitors inferior; refractory → retinoids, PDT. † Meatotomy/meatoplasty for meatal disease.

Disease	Steroid potency	Duration
AD / psoriasis / reactive arthritis	Low	Short (psoriasis ≤2 weeks)
Seborrhoeic dermatitis	Low	Initial severe phase only
Lichen planus	Ultrapotent	Until controlled

Disease	Steroid potency	Duration
Lichen sclerosus	Potent	3 months → maintenance

KEY POINT
Steroid strength tracks the scarring and malignant stakes of the disease; LS is the single exception to the short-course rule.

6 · Vesiculobullous and drug eruptions

Feature	EM minor	SJS (EM major)	Fixed drug eruption (FDE)
Trigger	**HSV** (10–14 days before)	**Drugs**: sulfonamides, NSAIDs, penicillins, cephalosporins, anticonvulsants	Sulfonamides, NSAIDs, tetracyclines, barbiturates, carbamazepine; rarely finasteride, tadalafil, fluconazole
Lesion	Symmetric **targets**; palms, soles, mouth	Prodrome → macules → blisters → necrosis; genital ~**60%**	Solitary violaceous plaque with halo, often glans
Course	Self-limited, recurs	4–6 weeks; mortality to ~**30%** (TEN overlap)	**Recurs at same site within 24 h** of re-exposure
Sequelae	None	Meatal stenosis, synechiae, strictures	Brown pigmentation
Treatment	Antihistamine; **suppressive aciclovir prevents**	See algorithm	Stop drug

ALGORITHM
SJS: stop offending drug → burn-type supportive care → ICU / burn unit (lowers mortality) → multispecialty wound care; limited data: cyclosporine 3–5 mg/kg/day, etanercept, plasmapheresis, IVIG; systemic steroids controversial

KEY POINT
The same drug class can cause a trivial FDE or a lethal SJS: extent and systemic illness separate them. Genital FDE mimics herpes — "same spot each time" is the clue.

† **Autoimmune blistering**: pemphigus vulgaris — intraepidermal, flaccid, Nikolsky +; pemphigoid — subepidermal, tense; mucous membrane pemphigoid scars (meatal stenosis). Perilesional biopsy + DIF.

7 · Noninfectious genital ulcers

Genital ulcer is infectious until proven otherwise → *Ch 29*.

Behçet disease (BD)

- Relapsing vasculitis; **HLA-B51**; highest in Turkey; **males more severe**. Oral + genital ulcers + uveitis; epididymitis, thrombophlebitis, **pulmonary artery aneurysm**.
- Oral ulcer most sensitive; **genital ulcer most specific**. No confirmatory test.

International Criteria for BD (2014) — ≥4 points	Points
Ocular lesions; oral aphthosis; genital aphthosis	2 each
Skin lesions; CNS; vascular; positive pathergy (optional)	1 each

Treatment: colchicine, dapsone, steroids, cyclosporine, **anti-TNF**.

Pyoderma gangrenosum (PG) and factitial

- PG: with IBD, arthritis, myeloproliferative disease, HIV. Painful ulcer, **undermined violaceous border**, pathergy. Biopsy edge (neutrophils). Systemic + local steroids ± cyclosporine, anti-TNF, IVIG. † Avoid debridement.
- **Factitial dermatitis**: oddly angulated erosions; psychiatric association.

8 · Infections

Balanitis and balanoposthitis

- Children: bacterial. Adults: hygiene, trauma, irritants, *Candida*, bacteria, viruses.

> **ALGORITHM**
> Remove irritants → hygiene → topical antibiotic/antifungal → short low-potency steroid → **failure: biopsy** (neoplasia, Zoon, psoriasis, HPV, HSV) → recurrent with phimosis: **circumcision**

Type	Clue	Management
Candidal	Satellite lesions; diabetes	Topical imidazole; check glucose
Irritant/allergic	Products, soaps	Avoidance
Circinate	Urethritis, HLA-B27	Low-potency steroid
Zoon (plasma cell)	Shiny moist red plaque, elderly uncircumcised	Biopsy → circumcision
LS / BXO	White sclerotic glans, phimosis	Potent steroid; circumcision
PEKMB	Mica-like scaly plaque, adult-circumcised	Biopsy; excision

Soft-tissue infection by depth

Entity	Level / organism	Sign	Management
Folliculitis	Follicle; *S. aureus*; hot tub → *Pseudomonas*	Perifollicular pustules	Hygiene; culture if refractory
Furuncle	Deep follicle abscess; *S. aureus*	Tender boil	I&D; MRSA cover; nasal mupirocin if recurrent
Erysipelas	Upper dermis + lymphatics; *S. pyogenes*	Sharp border	Antibiotics
Cellulitis	Deep dermis + subcutis; mixed in diabetes	Indistinct border	Systemic antibiotics ± MRSA; **mark the border**
Fournier gangrene	Fascia; polymicrobial	Pain out of proportion, grey skin, crepitus	Emergency debridement → *Ch 26*

> **KEY POINT**
> Fournier gangrene is fascial — skin shows a fraction of the necrosis; testes spared (own blood supply). Imaging (CT best for gas) must never delay surgery. Debride to bleeding tissue; second look at 24–48 h.

- **Ecthyma gangrenosum**: *P. aeruginosa* bacteraemia in immunocompromised; haemorrhagic bulla → black eschar. Immediate IV antipseudomonals ± debridement.
- **Genital bite**: *Eikenella corrodens*; amoxicillin–clavulanate 875/125 mg bd + irrigation, debridement.

Hidradenitis suppurativa (HS)

- Apocrine-bearing skin after puberty; female, Black ancestry; familial up to 41%. **Follicle-centred**: plugging → rupture → abscesses, **sinus tracts**, scars.
- Complications: fistulae to urethra/bladder/rectum, lymphoedema, amyloidosis, **SCC in chronic scars**.

† Hurley stage	Features	Mainstay
I	Abscess(es), no tracts/scarring	Topical clindamycin, deroofing
II	Recurrent abscesses, tracts, scarring; separated	Oral tetracycline / clindamycin + rifampicin; deroofing; adalimumab if refractory
III	Diffuse interconnected tracts	Adalimumab + **wide excision**, graft/flap

- Smoking cessation, weight loss. Topical clindamycin = oral tetracycline (RCT); dapsone, acitretin, metformin; spironolactone (women), 5-ARI (men); **adalimumab 40 mg weekly**. **Avoid lithium**.
- **Repeated incision and drainage discouraged** — tract recurs; wide excision with grafting or CO_2 laser for chronic disease.

Red groin rash

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
Organism	*Candida*	Dermatophytes	*Corynebacterium minutissimum*	None
Scrotum	**Involved**	**Spared**	—	May be involved
Look	Macerated; **satellite lesions**	Scaly advancing border, central clearing	Red-brown scaly patch	Non-scaling plaque
Test	KOH: yeast + pseudohyphae	KOH: hyphae	Wood's: **coral-red**	KOH negative
Treat	Topical imidazole **≥2 weeks**; fluconazole 150 mg occasionally; test for diabetes	Topical azole/allylamine; deep → terbinafine 250 mg/day 1 week; no steroids	Antibacterial soap, clindamycin 1%, miconazole; oral erythromycin if severe	Low-potency steroid

> **KEY POINT**
> Azoles cover yeast and dermatophyte (use when unsure); nystatin misses tinea; allylamines weaker on *Candida*. Negative KOH from an untreated lesion → rethink the diagnosis.

Infestations

Feature	Pediculosis pubis	Scabies
Sign	Lice/nits on hair; eyelashes	Nocturnal itch; **serpiginous burrows**; crusted in immunosuppressed
First-line	Permethrin 5% to hair-bearing areas, **repeat at 1 week**	Permethrin 5% whole body overnight, **repeat at 1 week**, treat household
Alternatives	Oral ivermectin; malathion 0.5%	Ivermectin 200 µg/kg ×2; lindane disfavoured (CNS)

Feature	Pediculosis pubis	Scabies
Extra	Screen for gonorrhoea/chlamydia	Postscabies glans nodules → intralesional triamcinolone

9 · Premalignant and malignant lesions

- P1 Premalignant conditions in urology (including premalignant lesions of penis) — *PeIN HPV vs non-HPV; Queyrat, Bowen, bowenoid papulosis; Buschke–Löwenstein (no RT); glans therapy (5-FU, imiquimod, laser, resurfacing)*
- Other risks: immunosuppression, radiation, arsenic. Bowen: seventh decade. Bowenoid papulosis: third decade, **regresses**; female partners need cervical screening. Biopsy **several areas**.

Lesion	Treatment
SCCis — accessible (scrotum)	Excision with **5-mm margin** (favoured)
SCCis — glans	Mohs, laser, cryo; topical 5-FU/imiquimod (higher recurrence)
Bowenoid papulosis	Observe in reliable young man; or 5-FU 0.5%, imiquimod, cryo, laser — relapse common
PEKMB	Biopsy (exclude verrucous/SCC); excision/ablation, 5-FU; refractory → glans resurfacing

KEY POINT

HPV mapping: 6/11 → condyloma and verrucous carcinoma (locally destructive, rarely metastasises); 16 → bowenoid papulosis and SCCis.

Tumour	Key features	Treatment
Basal cell carcinoma	Rare; pearly papule, telangiectasia	Excision/Mohs; advanced → vismodegib, sonidegib
Kaposi sarcoma	HHV-8; classic, AIDS (genital ~3%), transplant; glans lesion may block meatus	Restore immunity (ART; calcineurin → mTOR inhibitor); solitary → excision/laser/intralesional vinblastine; regional → RT 15–30 Gy; disseminated → chemo
Melanoma	<1% of penile cancers	Wide excision/partial penectomy; advanced → BRAF/MEK, anti-CTLA-4, anti-PD-1
Extramammary Paget	Red plaque at apocrine sites; CEA/EMA +; underlying carcinoma **10–30%** (urethra, prostate, bladder, rectum)	**Mandatory search for internal malignancy**; excision/Mohs; imiquimod, 5-FU, RT
Mycosis fungoides	Pruritic patches → plaques → tumours; Sézary if blood	Ultrapotent steroid, nitrogen mustard, phototherapy, electron beam

10 · Benign male genital lesions

Lesion	Site / look	Mimic	Management
Pearly penile papules	Rows on corona; 14–48% young men; angiofibromas	Condyloma (no HPV)	Reassure; laser/cryo if cosmetic
Fordyce spots	Pin-sized shaft papules (ectopic sebaceous)	Warts	Reassure

Lesion	Site / look	Mimic	Management
Angiokeratoma of Fordyce	1–2 mm red-purple scrotal papules; bleed	Fabry disease	Observe; laser if bleeding
Zoon balanitis	Shiny moist red glans plaque; elderly uncircumcised; plasma cells	Queyrat, EPD	**Biopsy → circumcision (curative)**
Sclerosing lymphangitis	Coronal cord after vigorous sex	—	Self-limiting
Median raphe cyst	Ventral, near glans; no urethral link	—	Excision

KEY POINT

Zoon balanitis vs erythroplasia of Queyrat: both red glans plaques in uncircumcised men — one benign and cured by circumcision, the other SCCis. A persistent red glans plaque always needs biopsy.

11 · Pigmentary disorders

- Vitiligo: Wood's lamp accentuates; NB-UVB / excimer for genital disease. † Not premalignant; unlike LS, normal texture, no atrophy/scarring.

31 Tuberculosis and Parasitic Infections of the Genitourinary Tract

1 · Urogenital tuberculosis — organism and pathogenesis

- P3 P2 Renal and genitourinary TB: etiopathogenesis, presentation, diagnosis and management — *MTBC, granuloma/IFN-γ, latency and reactivation, routes, renal sequence to autonephrectomy, site frequencies*
- **UG TB** preferred term (urinary tract involved far more than genital). Second "great imitator" after syphilis. ***M. bovis* (and BCG) innately pyrazinamide-resistant**; *M. africanum* in West Africa.
- **15–20%** of pulmonary TB in developing countries; latent 90–95%, lifetime reactivation **5–15%**; latency up to decades (46 years) → UG TB often without active lung disease.
- Before cellular immunity (up to 12 weeks) bacilli spread via hilar nodes and blood. Granuloma (epithelioid cells, Langhans giant cells, lymphocyte cuff) maintained by **IFN-γ** (also IL-2, TNF-α). Reactivation: old age, renal failure, diabetes, malnutrition, HIV, immunosuppression.
- Isolated UG TB generally **not infectious** — except skin inoculation or aerosol-generating procedures (theatre).
- Ascending: intravesical BCG → UG TB in **~1%**. Contiguous: spine → psoas → kidney; gut TB → enterorenal/enterovesical fistula.

> **KEY POINT**
> UG TB is a blood-borne disease landing in the two best-perfused UG organs — kidney and epididymis. Haematogenous seeding gives parenchymal disease; descending urinary spread gives luminal disease (distal ureteric stricture, cystitis). Healing by fibrosis turns an infection into an obstruction.

2 · Clinical features and pathology by site

- Indolent; repeatedly treated as bacterial UTI or investigated for cancer. Constitutional symptoms **< 20%**; **LUTS dominate**; haematuria/flank pain ~⅓; colic <10%. **Hallmark: sterile pyuria** (>90%).

Site	Frequency / route	Signature lesion
Kidney	**80%**, haematogenous	Caseating cavities → autonephrectomy (up to ⅓); "putty" kidney; ESRF ~7%; pelvic squamous metaplasia → SCC risk
Ureter	Descending	Stricture at VUJ; beaded corkscrew; may obstruct and reflux. **Stricture obstruction is an important cause of renal failure**
Bladder	Descending	Starts at orifices; dome worst, trigone spared → after ~1 year <100 mL; "thimble" <20 mL
Epididymis	10–55%; second commonest; bilateral ~⅓	Starts in **globus minor (tail)**; posterior scrotal sinus in up to half; spreads to testis
Vas	Contiguous	Thickened, beaded
Testis	From epididymis; isolated rare	Hard tumour-like mass; hydrocele ~5%
Prostate	22–49%	Calcified hard gland; "prostatitis" with partial quinolone response then relapse (quinolones active on MTBC)

Site	Frequency / route	Signature lesion
Seminal vesicles	Via vas/ejaculatory ducts	Low-volume ejaculate, azoospermia, haemospermia — infertility may be first symptom
Urethra	1.9–4.5%	With prostatic disease; urethroscrotal fistula
Penis	Exceedingly rare	Ulcer; **papulonecrotic tuberculid** — hypersensitivity, culture/PCR negative, diagnosed by response to treatment

- Kidney without imaging change: granulomatous interstitial nephritis (reversible renal failure), glomerulonephritis, amyloidosis, miliary TB.

3 · Diagnosis

- P3 Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management — *3–5 early-morning urines, ZN, LJ/MGIT culture (gold standard, monitors treatment), GeneXpert, LF-LAM, TST/IGRA, imaging signs, cystoscopy, algorithm*

ALGORITHM

Hierarchy: culture or MTBC DNA positive → diagnosis. Negative → caseating granulomas on tissue in right context. Negative → compatible picture + exposure + response to empirical treatment. Always look for coexisting pulmonary TB (up to one-fifth): chest radiograph in every patient.

- Culture sensitivity 10–90% (intermittent shedding); ZN smear **< 50%**. Acidic urine inhibits mycobacteria → culture promptly.
- **GeneXpert**: >90% of rifampicin-resistant strains are INH-resistant (MDR surrogate). WHO: initial work-up of all TB; USA: adjunct — negative NAAT does not exclude. **Cannot monitor treatment** (dead bacilli shed DNA).
- Biopsy consistent in only ⅕–⅓; epididymal FNAC diagnostic in ~⅔. Cystoscopy: ulcers mimic cancer; **"golf-hole" orifice** → image upper tract.
- **TST** cut-offs: **≥ 5 mm** contacts/immunosuppressed; **≥ 10 mm** immigrants, high-risk workers, drug injectors; **≥ 15 mm** no risk factors. False negative 10–25%. **IGRA**: no BCG false positives; preferred screen for those from endemic countries. Both = exposure only.

Modality	Signs / role
KUB	Calcification 24–44%; **lobar "putty"/"cement" kidney** pathognomonic of end-stage; ureteric calcification **intraluminal** (schistosomiasis: mural)
IVU	**Moth-eaten calyx**; **phantom calyx**; **Kerr's kink**; **pipe-stem ureter**; beaded corkscrew ureter; contracted bladder. Findings mostly **obstructive**
CT urography	Standard; granuloma may mimic RCC; less sensitive than IVU for earliest papillary change
US	**Serial monitoring of hydronephrosis during treatment**
MRI	DWI separates pyonephrosis from hydronephrosis, grades fibrosis; avoid gadolinium in renal failure

KEY POINT

Imaging signs are fibrosis made visible: destruction (moth-eaten calyx, cavities) heals by scarring (phantom calyx, Kerr's kink, pipe-stem ureter, golf-hole orifice, thimble bladder). Obstruction is the commonest finding and must be watched during treatment.

4 · Medical treatment

- Multidrug: bacilli in different metabolic states; combinations prevent resistance. Baseline LFT, renal function, platelets, HIV; DOT. First-line drugs reach high urinary levels.

ALGORITHM

Standard 6-month short course (2HRZE / 4HR): Intensive 2 months — daily INH (H) + rifampicin (R) + pyrazinamide (Z) + ethambutol (E); stop E once fully susceptible. Continuation 4 months — HR daily (or thrice weekly by DOT; twice weekly no longer recommended) → eradicates persisters. Pyridoxine throughout.

Drug	Daily adult dose	Signature toxicity	Monitor / note
INH	5 mg/kg (300 mg)	Neuropathy, hepatitis	Pyridoxine 25–50 mg
Rifampicin	10 mg/kg (600 mg)	Hepatitis, flu-like, **drug interactions**	Antiretroviral/immunosuppressant levels; rifabutin with antiretrovirals (uveitis)
Pyrazinamide	Weight-banded 1000–2000 mg	Hyperuricaemia, arthralgia, hepatitis	*M. bovis* resistant
Ethambutol	Weight-banded 800–1600 mg	Optic neuritis (red–green)	Vision; **only non-hepatotoxic** first-line drug

- **Not for UG TB**: weekly rifapentine; 4-month rifapentine–moxifloxacin. Complicated disease: some give 12–24 months (or ≥6 months after surgery) — UG TB relapses more.
- **Monitor**: LFTs, vision, urine culture, **serial US** — healing fibrosis can worsen strictures/bladder contraction.
- **Steroids**: proven in meningitis and pericarditis; in UG TB anecdotal — consider if stricture worsens on treatment. Rifampicin increases steroid metabolism†.

5 · Surgical treatment

- ~Half need surgery; most now **reconstructive**; goal organ preservation. Indications: obstruction/abscess, non-functioning infected kidney, resistant hypertension, reconstruction.
- **Timing: 4–6 weeks after starting chemotherapy** (emergencies excepted).

ALGORITHM

Uraemia or sepsis → emergency decompression → retrograde double-J stent (preferred; succeeds in < half, 41%) → fails → antegrade stent → fails → percutaneous nephrostomy (PCN; loculated systems may need more than one) → definitive repair; unsalvageable kidney → nephrectomy.

- Avoid high contrast pressure (dissemination). Removing PCN without fixing the cause → tuberculous cutaneous fistula.
- **Nephrectomy — established**: non-functioning kidney with recalcitrant TB, or with resistant hypertension (improves ~⅔). **Contested**: routine removal of asymptomatic non-functioning kidney. Partial nephrectomy for polar lesion unresponsive to drugs.
- Technique: oblique extendable retroperitoneal incision (dense fibrosis); en bloc caseous perinephric tissue; **ligate artery and vein separately** (late AV fistula); ureter usually left; avoid pleura/peritoneum; laparoscopy increasingly preferred.

- **Strictures**: lower commonest; endoscopic results poorer (ischaemic fibrosis); best if short with residual lumen; early DJ for strictures arising on treatment; balloon dilatation often fails. **Image every stricture**; worsening → steroids; **no improvement after 6 weeks → open surgery**.

Site / gap	Procedure	Key point
PUJ, short	Dismembered pyeloplasty; flap if longer	Harder than congenital
PUJ unreconstructable	Ureterocalicostomy (lower pole)	Cover with capsule/omentum
Upper/mid ureter	Ureteroureterostomy; Davis intubated ureterotomy	—
Lower ureter	Excise → ureteroneocystostomy	Bladder mobilisation gains 2–3 cm
Moderate gap	Psoas hitch	Up to **5 cm**; needs good capacity; avoid genitofemoral/femoral nerves
Long gap	Boari flap ± hitch	**10–15 cm**; poor in contracted bladder
Multiple / recurrent	**Ileal ureter**	Native ureter unusable

KEY POINT
Trap: in TB the bladder itself may be contracted, ruling out psoas hitch and Boari flap — assess capacity first; ileal ureter ± augmentation is then the option.

- P2 Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male — *assessment, augmentation vs substitution, ureteric reimplantation, bowel complications, follow-up*
- Contracted bladder: operate for intolerable symptoms, capacity **< 100 mL**, after chemotherapy controls disease; thimble → orthotopic substitution. Urethra: bladder neck contracture → incision; fistula → suprapubic diversion, delayed repair.
- **Genital**: only after failed medical therapy. Testis spared → **epididymectomy** (start at globus minor, ligate vas — diseased distal to external ring, so ligate there); testis involved → scrotal orchidectomy. Obstructive azoospermia often irreversible → sperm retrieval + ART†.

6 · BCG-related infection, relapse, special situations

- **BCG UG TB** (~1%): granulomatous prostatitis, epididymo-orchitis, renal abscess, ureteric obstruction, BCG sepsis; Mantoux unhelpful. † Local cystitis → withhold BCG, quinolone; organ infection → INH + rifampicin (+ E) 3–6 months; **pyrazinamide useless**; sepsis → triple therapy + steroids, stop BCG permanently.
- **Relapse 6.3–22%** (pulmonary 2–6%), mean ~5 years; follow-up **10 years**; culture recurrences.

Situation	Change	Duration
MDR-TB (INH + R resistant)	Second-line; BPaL (bedaquiline + pretomanid + linezolid), not yet approved for extrapulmonary	> 18 months; 6–9 months with BPaL
XDR-TB	MDR + fluoroquinolone + bedaquiline/linezolid resistance; 5–6 drugs	≥ 2 years
Pregnancy	Treat promptly; HRE + pyridoxine; Z omitted in USA (WHO includes); breastfeeding allowed	9 months (without Z)

Situation	Change	Duration
HIV on effective ART	Standard daily; rifabutin with protease inhibitors	6 months (9 if not on ART)
Renal transplant	Rifamycin with drug-level monitoring, or rifamycin-free	≥ 18 months if rifamycin-free

- **HIV**: risk 15–20-fold; more disseminated, **less fibrosis, fewer strictures**, higher mortality. **Transplant**: allograft TB within 6 months, fever, normal imaging → **urine AFB culture**.

7 · Schistosomiasis (bilharzia)

Species	Distribution	Target / venous home	Snail
S. haematobium	Africa, Arabian Peninsula	Vesical/pelvic plexus — UG tract	***Bulinus***
S. mansoni	Africa, Middle East, S America	Mesenteric — liver, gut	*Biomphalaria*
S. japonicum	China, SE Asia	Mesenteric — liver, gut	*Oncomelania*

ALGORITHM

Fork-tailed cercariae penetrate intact skin in fresh water → schistosomulae → lungs → liver → paired adults (female in male's gynaecophoric canal) to vesical plexus (80–110 days; live 3–5 years) → eggs (~140 µm, **terminal spine**) cross bladder wall into urine → miracidia → *Bulinus* → cercariae.

- **Retained eggs cause disease**; adults evade immunity (tegument), don't multiply in humans. Children 5–15 heaviest burden; dams/irrigation spread it.
- **Egg granuloma** → polypoid masses (active) → fibrosis/calcification → **"sandy patches"** → obstruction, metaplasia, cancer. Hydroureter precedes hydronephrosis; tubular function lost before glomerular.
- Obstruction **distal** (intramural/juxtavesical); meatus ~1%. Hydroureter: **segmental**; **tonic** (reversible once relieved); **atonic** (fibrotic, failed).
- ***Salmonella*** shelters in tegument → chronic bacteriuria/bacteraemia; treat both infections.
- **Bladder cancer**: IARC **Group 1**; hyperplasia → squamous metaplasia → dysplasia → carcinoma; chronic inflammation, VEGF, bacterial nitrosamines alkylate DNA (p53). Younger (40–50), **SCC 60–90%**, adenocarcinoma 5–15%, posterior wall; MDA in Egypt shifted SCC → urothelial.
- **Acute**: swimmer's itch; **Katayama fever** 2–8 weeks (rare with *S. haematobium*). **Chronic**: from 8–12 weeks, **painless terminal haematuria**.
- Stages: active (polyps) → late chronic active (contracted bladder ~50 mL, trigone spared) → quiescent → chronic inactive (no viable eggs; 40–60% with obstruction first present here).
- **Female genital** (⅓–¾ of infected women): cervical/vaginal sandy patches, infertility, ?HIV risk. **Male**: haemospermia (eggs in semen before urine), funiculitis mistaken for tumour, infertility (often reversible).

KEY POINT

Trap: a negative urine egg count does not exclude severe disease — the most damaged kidneys belong to egg-negative, often asymptomatic, inactive-stage patients.

Test	Meaning	Limitation
Urine egg microscopy — **gold standard**	Active infection	Collect **09:00–15:00** (noon peak); negative first 8–12 weeks and in light/inactive disease
Rectal snip (squash prep)	Active/past; egg viability	Avoids bladder biopsy risks
Serology (FAST-ELISA + Western blot)	Exposure (>90%; from 4–6 weeks)	Persists — **never judges cure**
Circulating anodic/cathodic antigen	Live worms	Urine CCA poor for *S. haematobium*
PCR	Most sensitive	Impractical in field

- Imaging: US wall thickening, polyps, hydronephrosis; plain film **calcified bladder — "fetal head in the pelvis"**, mural ureteric calcification; IVU ureteric striations (earliest); VCUG reflux ~25%. Cystoscopy: sandy patches, polyps, ulcers, golf-hole/stenotic orifices, tumours.
- **Praziquantel** — only WHO drug: **40 mg/kg once**, or 20 mg/kg twice 6–8 h apart; cure/egg reduction 60–90%. Pregnancy FDA category B. Katayama: add steroids. Weak on schistosomulae → repeat after several weeks. Artemisinins act on schistosomulae — avoid where malaria co-endemic.
- Active polypoid obstruction resolves with PZQ alone (1–2 months; children 2–6 weeks); late fibrotic stenosis (up to 80%) → surgery.

Complication	Surgery
Distal / intramural stenosis	Antireflux reimplantation; balloon (restenosis common); meatoplasty for meatal stenosis
Long / multifocal stenosis	Boari flap; ileal ureter; ileal conduit; long-term nephrostomy
Contracted bladder	Ileocystoplasty or diversion, always with PZQ
Chronic deep ulcer	Partial cystectomy (fulguration fails)

8 · Filariasis

- *W. bancrofti* (>90%; *Culex* in urban India), *Brugia*. **Only *W. bancrofti* causes genital filariasis** (Brugia: leg below knee). Adults 5–7 years; microfilariae **nocturnally periodic**. *Wolbachia* endosymbiont killed by doxycycline. Needs years of exposure; genital disease rare before teens.
- **Acute adenolymphangitis**: **retrograde** lymphangitis (bacterial ascends). **Funiculoepididymitis**: recurrent painful cord; testis spared; often mistaken for tumour → needless orchidectomy.
- **Hydrocele** commonest chronic lesion (thick tunica, cholesterol/calcium, cord nodules). Elephantiasis late; lymph scrotum.
- Diagnosis: midnight Giemsa smear (minority positive); **circulating antigen** card (*W. bancrofti* only); IgG4 = exposure; US "**filarial dance sign**"; lymphoscintigraphy. "Burned-out" disease: tests negative.
- P3 Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria — *pyelolymphatic fistula; ether test, triglycerides, cystoscopic lateralisation, lymphoscintigraphy; MCT diet + DEC → pelvic instillation sclerotherapy → hilar lymphatic disconnection*

Drug	Action	Regimen / caution
DEC	Microfilaricidal, modest on adults	Drug of choice: 6 mg/kg/day (2 mg/kg tds); single day ≈ 12 days. **Contraindicated if onchocerciasis or *Loa loa* possible** (blindness, encephalopathy)
Albendazole	Micro- and macrofilaricidal	400 mg bd 21 days; single dose in MDA
Ivermectin	Microfilaricidal	150–400 µg/kg once; caution with *Loa*
Doxycycline	Anti-*Wolbachia*; kills adults	200 mg daily 4–8 weeks; improves lymphoedema/hydrocele; not in pregnancy/children

- Treat all microfilaraemic patients. MDA: albendazole + ivermectin (Africa, Loa/oncho co-endemic) or albendazole + DEC (incl. India); triple (IDA) more suppressive.
- **Hydrocele**: aspiration recurs → **excision of sac**; ligate leaking lymphatics. Funiculoepididymitis: excise nodules, preserve testis. Elephantiasis: reduction scrotoplasty; lymphadenectomy worsens drainage.
- Lymphoedema: hygiene, prevent bacterial infection, elevation, compression, physiotherapy.
- **Chyluria**: rupture of dilated retroperitoneal lymphatics into the collecting system → milky urine, fat/protein loss; intermittent, may remit; precedes elephantiasis. Exclude TB, tumour, trauma.
- Onchocerciasis ("hanging groin"): ivermectin then doxycycline; DEC blinds. Loiasis (haematuria ~30%).

9 · Hydatid disease and other parasites

- P1 Hydatid disease of kidney — *kidney third site (2–3%); hydatiduria; water-lily sign, WHO stages; albendazole 400 mg bd 1–6 months; no aspiration; scolicide only if non-communicating; nephron-sparing surgery; PAIR not for kidney*
- Slow growth (1–2 cm/year); function usually preserved; serology adjunctive (60–90%). Perioperative albendazole (± PZQ) reduces seeding. Pack field with hypertonic (20%) saline swabs; remove intact (pericystectomy) or partial nephrectomy; nephrectomy only for destroyed kidney; anaesthetist ready for anaphylaxis.
- **Enterobiasis**: ectopic migration in women (vulval granulomas, salpingitis); rarely epididymis. Albendazole 400 mg or mebendazole 100 mg once, repeat at 2 weeks; treat household.
- **Amoebiasis**: renal involvement, painful genital ulcers. Tinidazole/metronidazole, then paromomycin or iodoquinol.
- Trichomoniasis → *Ch 29*; fungal UTI → *Ch 26*.

★ Putting it together: infection decisions

A · UTI: drug matches site

Situation	Choice	Principle
Uncomplicated cystitis	Nitrofurantoin 5 days, fosfomycin 3 g once, TMP–SMX 3 days (resistance <20%), pivmecillinam	Narrow, urine-confined, flora-sparing
Pyelonephritis, complicated, men	Tissue-penetrating (quinolone, cephalosporin; carbapenem if resistant)	Nitrofurantoin/fosfomycin don't reach tissue
ESBL	Fosfomycin/nitrofurantoin for cystitis; carbapenem if invasive	Co-resistance to quinolones, aminoglycosides
Asymptomatic bacteriuria	Treat only in pregnancy, before mucosa-breaching surgery	Else harm

B · Beyond antibiotics

ALGORITHM

Image (US/CT): no response by 72 h, sepsis, obstruction/stone, gas, febrile men, recurrent pyelonephritis.

Drain any obstructed infected kidney (stent/nephrostomy); renal abscess ≤3 cm → antibiotics alone; 3–5 cm variable; >5 cm → percutaneous drainage. Treat the stone later.

Emphysematous pyelonephritis (CT class): gas within kidney → antibiotics + drainage; beyond kidney, poor prognosis → nephrectomy (→ *Ch 26*).

Fournier: antibiotics, **urgent repeated radical debridement**; scores never delay surgery.

C · Recurrent UTI in women

ALGORITHM

Culture-confirm → behaviour, hydration → vaginal oestrogen (postmenopausal) → methenamine, cranberry, D-mannose, immunostimulants → self-start or post-coital antibiotic → continuous low-dose prophylaxis.

D · Pelvic pain syndromes compared

	CP/CPPS	IC/BPS
Core	Male pelvic pain >3 months, no infection (NIH III)	Filling-related bladder pain + LUTS, no other cause
Phenotyping	UPOINT; NIH-CPSI	Hunner vs non-Hunner (cystoscopy)
Treatment	By domain: α-blocker, antibiotic only if infection, pelvic-floor PT, neuromodulators, psychology	Behavioural → oral (amitriptyline, PPS) → intravesical → Hunner fulguration/injection → BTX-A, neuromodulation → diversion last

E · GU tuberculosis in four lines

ALGORITHM

Suspect: sterile pyuria, contracted bladder, beaded vas, epididymal mass.

Confirm: three early-morning urines: AFB, culture, GeneXpert.

Treat: 2 months HRZE, then 4 months HR; watch for ureteric stricture.

Operate after ≥4–6 weeks of drugs: nephrectomy (non-functioning), reimplantation (stricture), augmentation (contracted bladder) (→ *Ch 31*).